

A Survey on Digital Twin for Industrial Internet of Things: Applications, Technologies and Tools

Hansong Xu, Jun Wu, Qianqian Pan, Xinpeng Guan, and Mohsen Guizani

Abstract—Digital twin for the industrial Internet of Things (DT-IIoT) creates a high-fidelity, fine-grained, low-cost digital replica of the cyber-physical integrated Internet for industry. Powered by artificial intelligence (AI) and security technologies, DT-IIoT provides advanced features such as real-time monitoring, predictive maintenance, remote diagnostics, and rapid response for smart IIoT systems. A systematic review of key enabling technologies such as digital twin, AI, and blockchain is essential to develop DT-IIoT and reveal pitfalls. This paper reviews the preliminaries, real-world applications, architectures and models of digital twin-driven IIoT. In addition, advanced technologies for intelligent and secure DT-IIoT are investigated, including state-of-the-art AI solutions such as transfer learning and federated learning, as well as blockchain-based security solutions. Moreover, software tools for high-fidelity digital twin modeling are proposed. A case study on reinforcement learning-based integrated-control, communication, and computing (3C) design is developed to demonstrate the AI-driven intelligent DT-IIoT. Finally, this paper outlines the prospective applications, challenges, and integrations with ABCDE (i.e., AI, Blockchain, cloud computing, big data, edge computing) as the future directions.

Index Terms—Digital Twin, Industrial Internet of Things, Artificial Intelligence, Blockchain, Integrated Design.

I. INTRODUCTION

DT-IIoT leverages digital twins to create digital systems that integrate control, communication, and computing (3C) of the cyber-physical integrated Internet of industry [1]. The control subsystems in IIoT systems represent industrial control systems such as supervisory control and data acquisition (SCADA), etc.. The communication subsystem transmits sensing data and actuating commands between sensors, controllers, and actuators. The computational subsystem provides computational resources and computes the actuation signals. Note

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that most existing digital twin studies create a “virtual scene” [2], [3] for physical entities and systems. It is unclear how digital twins can contribute to the development of digital testbeds in a high-fidelity, fine-grained, and cost-effective manner for intelligent and secure cyber-physical integration internet of industrial systems. Furthermore, comprehensive digital twins pursue integrated resource allocation and management solutions, such as holistic integrated 3C design, which remain inadequate in existing research. Therefore, a comprehensive survey of current research advances, state-of-the-art intelligence and security technologies, software tools, and DT-IIoT case studies can help address the above issues.

Digital twin technologies have received extensive academic attention and in-depth research, including computing [4], automation [5], [6], system simulation [7], and others [8]. In particular, digital twin is considered as a key enabling technology for smart manufacturing systems, which integrates novel information technologies such as AI and blockchain [9]. Specifically, the digital twin uses physical models and historical data to reflect the entire lifecycle of physical entities and systems. The bi-directional real-time interaction between the physical and virtual domains enables the digital twin to record, monitor, and predict the trajectory of physical entities and systems, and to conduct optimal resource allocations.

In addition, the digital twin was first proposed and received attention from industry organizations such as National Aeronautics and Space Administration (NASA) and continues to play a key strategic role in many industry organizations [25], [26]. The digital twin was listed by Gartner as one of the top 10 strategic technology trends for 2019 alongside AI and blockchain [27]. The U.S. Air Force Research Laboratory (AFRL) first proposed the creation of a digital twin for fighter jets in 2011 to assess and maintain the status of fighter jets. Subsequently, General Electric (GE) proposed the creation of a digital twin for aero engines to analyze and predict engine conditions and optimize management. In 2020, the Industrial Internet Consortium (IIC) released a white paper aimed at promoting the development and application of digital twins in industry [2], [3]. In addition, the University of Cambridge has established the Centre for Digital Built Britain to lead the “National Digital Twin Project” [28]. The German government is funding 32 million euros in 2020 for the creation of a Connected City Twin, an urban data platform and digital twin, across multiple cities (i.e. Hamburg, Leipzig and Munich) for integrated urban development [29].

However, existing digital twin research faces three challenges that are the key motivations of this paper. First, the current digital twins, which do not integrate all 3C subsystems

TABLE I. An Overview of Selected Surveys in Digital Twin and IIoT Over Past Five Years

Year	Reference	Remark	Applications	Intelligence	Security	Tools
2018	Tao et al. [10]–[12]	Digital twin-driven product design, manufacturing and service in the era of big data	✓			
	Kritzing et al. [13]	A classification study based on the level of digital twin integration in manufacturing	✓			
2019	Tao et al. [14]	A comprehensive review of the developments and challenges in the application of digital twin technology in industry	✓			
	Cimino et al. [15]	A review of digital twin services and applications in manufacturing	✓			
	Qi et al. [16]	Review of tools for digital twin modeling, data management, services, and connectivity	✓			✓
2020	Minerva et al. [17]	A comprehensive review of digital twin for IoT	✓			
	He et al. [18]	Intelligent digital twin design review for sustainable manufacturing		✓		
	Liu et al. [19] Lu et al. [20]	A comprehensive review on the digital twin definitions, technologies and applications	✓	✓		
2021	Wu et al. [21]	A comprehensive survey on technologies, applications and issues of digital twin network	✓			
2022	Mihai et al. [22]	A comprehensive survey on technologies, use cases and prospects of digital twin and case studies	✓	✓		
	Alcaraz et al. [23]	Review of potential threats to the digital twin paradigm and corresponding security approaches			✓	
2023	Wang et al. [24]	Study distributed Internet of digital twins architectures and methods for classifying and defending against security and privacy threats			✓	
	Our paper	A comprehensive survey on the applications, intelligence and security technologies, and tools of the DT-IIoT	✓	✓	✓	✓

of IIoT systems, are simple “digital abstraction” of the physical domain. They are coarse-grained and have no practical operability. Second, current digital twin systems are rarely integrated with AI, which makes them incapable of effective analysis, prediction, and automation with the vast amount of data available from the Internet of Things (IoT). Third, synchronization between digital twins and physical systems lacks a secure and trusted environment, which reduces the performance and security of the IIoT.

To address the above challenges, in this paper, we propose a “human-machine-object” architecture to elucidate the interactions between humans, digital twins, and internet for industry systems in Fig. 2. Then, we study the integrated-3C resource allocation methodologies involving parameters such as communication protocols, network topologies, control models, and system configurations. It is impractical to derive an accurate and comprehensive mathematical model for complex IIoT systems. Specifically, the integrated-3C design is complicated by considering the mechanical and electrical model param-

eters of the control subsystem, the physical and MAC layer parameters of the communication subsystem, and the storage and scheduling parameters of the computing subsystem. In addition, the interactions between the 3C subsystems must be considered in the integrated-3C design, which makes the model-based integrated 3C design impractical.

In addition, we investigated software tools for high-fidelity digital twin modeling and conducted a case study on integrated-3C design through model-free reinforcement learning in DT-IIoT. We used a representative continuous stirred tank reactor (CSTR) as an example. The reinforcement learning agent observes the system and adjusts actions (i.e. parameters from the 3C subsystem) according to the system state with little or no prior knowledge. The goal is to learn by trial and error an optimal strategy for assigning actions to the digital twin of the CSTR in different states.

Moreover, we investigate blockchain-based security solutions for smart IIoT systems that are subject to ubiquitous cyber attacks. The attacks are targeted to interrupt normal

TABLE II. Table of Abbreviations

Acronym	Definition
DT-IIoT	Digital Twin for Industrial Internet of Things
AI	Artificial Intelligence
AFRL	Air Force Research Laboratory
NASA	National Aeronautics and Space Administration
HMI	Human-Machine Interface
3C	Control, Communication and Computing
SCADA	Supervisory Control and Data Acquisition
CSTR	Continuous Stirred Tank Reactor
PLC	Programmable Logic Controller
IBVS	Image-based Visual Servoing
LSTM-ED	Long-Term Short-Term Memory Encoder-Decoder
IDT-SDVN	Digital Twin-based Software-Defined Vehicular Networks
UAV	Unmanned Aerial Vehicle
SVM	Support Vector Machine
IIoT	Industrial Internet of Things
IoT	Internet of Things
IoV	Internet of Vehicles
IIC	Industrial Internet Consortium
GE	General Electric Company
DQN	Deep Q-Network
TCN	Temporal Convolutional Neural Network
ODE	Ordinary Differential Equation
DITEN	Digital Twin Edge Networks
DoS	Denial of Service
MITM	Man-in-the-Middle
PoW	Proof of Work
PoS	Proof of Stake
PBFT	Practical Byzantine Fault-Tolerant
HIL	Hardware-in-the-Loop
CORE	Common Open Research Emulator
CoAP	Constrained Application Protocol
EMANE	Mobile Ad Hoc Network Emulator
MQTT	Message Queuing Telemetry Transport
WCPS	Wireless Cyber-Physical Simulator
ELK	Elasticsearch, Logstash, Kibana
MAE	Mean Absolute Error
MDP	Markov Decision Process
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
WSAN	Wireless Sensor and Actuator Network
ABCDE	AI, Blockchain, Cloud computing, big Data, Edge computing
HMI	Human Machine Interface

control operations. For example, attackers use a Stuxnet worm to control a programmable logic controller (PLC) that remotely controls vehicles in an IIoT system and sends false messages to roadside units, vehicles, etc. [30]. In the presence of untrusted and malicious physical entities (e.g., malicious controllers, actuators, etc.), there is an urgent need to establish a secure environment for integrated resource allocation. Blockchain provides a distributed trust solution for multi-party data interactions. Existing research proves that blockchain can effectively guarantee data integrity and privacy, which is sufficient to solve the trust problem of overall resource allocation across domains [31], [32]. The abbreviations used in this paper are listed in the Table II.

A. Related Work

We compare existing survey papers on digital twins from architecture and models, functionality, intelligence, and security

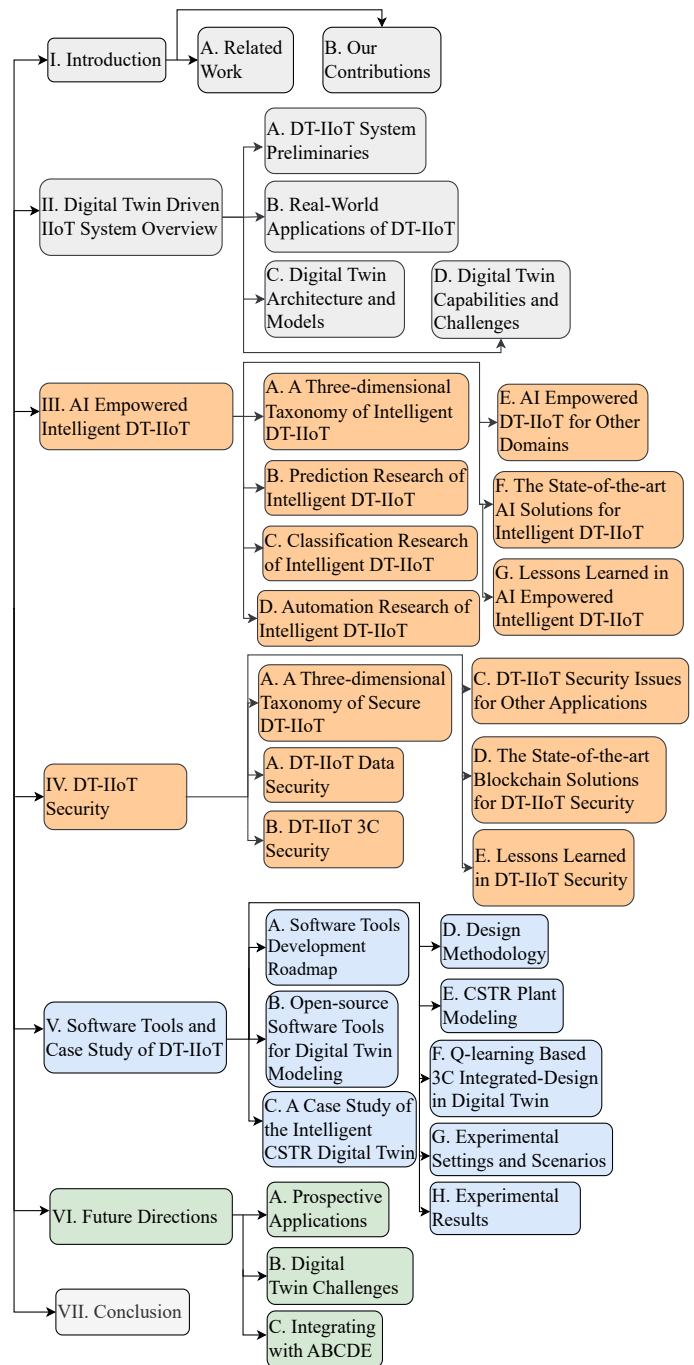


Fig. 1. The Paper Organization

perspectives, as shown in Table I. In the past five years, there have been many survey papers on digital twins. However, there are few survey papers that comprehensively investigate the digital twin of IIoT. We provide an overview and comparison of selected surveys in the field of digital twin and IIoT. DT-IIoT has distinctive features, such as stringent performance requirements, synthesized control, communication, computing, and critical infrastructure applications, which distinguish it from the digital twin of the general concept of IoT.

Unlike existing survey papers, this paper provides extensive coverage of the integration of AI techniques and security

solutions for IIoT digital twins from the perspective of architecture and models, functionality, intelligence, and security. This paper investigates not only classical machine learning solutions for prediction, classification, and automation problems for the IIoT but also state-of-the-art AI solutions such as transfer learning and federated learning. In addition, this paper investigates data security and control security issues for DT-IIoT, as well as state-of-the-art blockchain-based security solutions. We study software tools and conduct a case study on AI-driven intelligent DT-IIoT, including validation of CSTR process control systems and integrated control and computational resource allocation based on reinforcement learning.

From a digital twin research perspective, existing research efforts can be divided into three research categories: (i). Digital twin for industrial manufacturing systems, (ii). Digital twin-enabled communication networks, and (iii). Digital twin-based applications.

For category (i), Tao *et al.* combined the digital twin technology with the full lifecycle of industrial manufacturing, including design, manufacturing, and service [10], [11]. They proposed a five-dimensional digital twin model [14], [33], and workshop applications based on digital twin [34], [35]. Besides, Shao *et al.* carried out digital twin simulation research for industrial manufacturing from the perspective of simulation and system construction [36]–[38].

For category (ii), Zhang *et al.* applied digital twin technology to varieties of communication networks, such as the IoT [21], [32], [39], [40], edge network [41], air-to-ground network [42] to enable the dynamic representation, situational awareness, and automated management.

For category (iii), the digital twin has been adopted in many domains recently. For example, Wagg *et al.* applied digital twin technology to engineering dynamics applications, including structural dynamic applications [43], active vibration control [44], engineering dynamic model verification and validation [45], [46], structural health monitoring [47], and so on. Besides, El Saddik *et al.* applied digital twin to the fields of multimedia and health. They studied the digital twin coaches and health equipment interconnection [48]–[50], and promoted the establishment of a digital twin architecture for health applications standard (ISO/IEEE 11073) [51]. Moreover, Taylor *et al.* researched the application of digital twin technology in civil and environmental engineering, such as urban infrastructure management based on digital twin [52]–[54].

It is worth noting that DT-IIoT is a combination and extension of the three research categories mentioned above. Based on the above survey, we make the following observations:

- The digital twin continues to be a trending research topic and has been adopted into a wider range of application areas such as transportation, energy, etc. rather than the initial industrial applications.
- The digital twin becomes a ubiquitous technology that can be integrated with other trending technologies, such as big data, AI, and so on.
- The recent development of smart IIoT systems that require integrated sensing, modifying, analyzing, re-

healing, and thinking makes the digital twin a valuable and indispensable technology.

- There is a wealth of research work on the functionality of digital twins. However, the intelligence and security of the digital twin have not been sufficiently investigated, and this needs further investigation.
- Finally, there are no papers that comprehensively survey DT-IIoT with state-of-the-art AI, cybersecurity technologies, software tools, and case studies.

B. Our Contributions

The contributions of this paper are summarized as follows:

- We systematically survey the research efforts on the digital twin for the smart cyber-physical integrated internet for industry systems. We present the “human-machine-object” view of the intelligent and secure digital twin for smart industrial IoT systems.
- We investigate the research papers that synthesize AI and the digital twin. We discover the critical roles of the digital twin play in connecting human intelligence and AI machine intelligence. We present state-of-the-art AI solutions for digital twin, such as transfer learning and federated learning.
- We discuss the security issues of the digital twin from the data plane and the 3C plane. We then present the blockchain-enabled data management network for digital twin security.
- We review the software tools for high-fidelity digital twin modeling and carry out a case study on the integrated-3C design for the smart industrial system through reinforcement learning.
- We outline the future research directions of the digital twin, including the prospective applications, such as the “virtual-cyber-physical” integrated resource management and integrated system security, digital twin expectations, and integration with AI, blockchain, cloud computing, big data, edge computing (ABCDE).

We use Fig. 1 to summarize the organization of this paper. The remainder of this paper is organized as follows: we conduct an extensive review of the research papers on the digital twin in Section II. In Section III, we investigate the research efforts that integrate AI and digital twin, as well as state-of-the-art AI algorithms, such as transfer learning and federated learning. In Section IV, we investigate the research efforts on the security of AI-enabled digital twin and state-of-the-art blockchain solutions. Besides, we review the software tools and carry out a case study on the intelligent DT-IIoT in Section V. Finally, we outline the future research directions in Section VI and conclude this paper in Section VII.

II. DIGITAL TWIN DRIVEN IIoT SYSTEM OVERVIEW

In this section, we provide an overview of existing digital twin research in three areas, including preliminaries, real-world applications, and digital twin architectures and models. Then, we conclude this section with the characteristics and challenges of the digital twin.

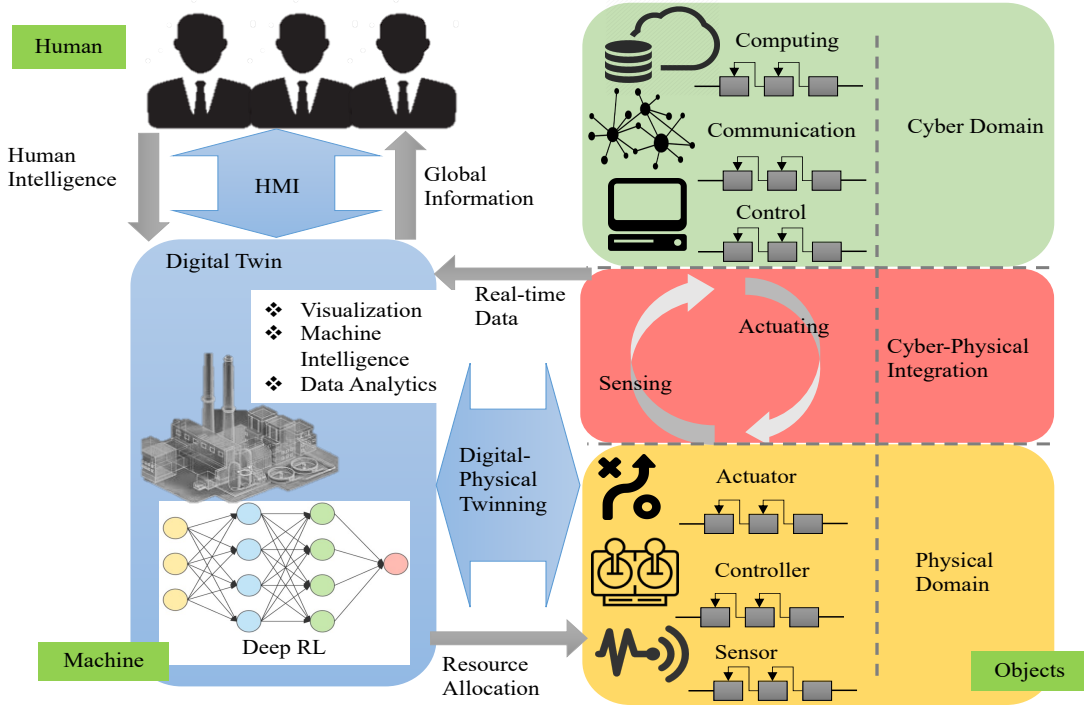


Fig. 2. The “human-machine-object” View of the Intelligent and Secure Digital Twin

A. DT-IIoT System Preliminaries

The smart IIoT system is an integrated system that synthesizes cyber operations and physical processes. Cyber operations include communication, computation, and control operations. Physical processes typically vary by application area, such as industry, transportation, and so on. The key feature of smart IIoT systems is the integration of 3C and physical components that enable traditional embedded systems such as industrial, energy, and transportation to exchange information and share resources. Smart IIoT systems require AI technologies for predictive and automation capabilities to further improve system functionality, intelligence, and efficiency. Digital twins create a fine-grained digital replica of smart IIoT systems, including cyber and physical components, data and processes. The digital twin runs AI algorithms for high-fidelity experiments and interacts with the smart IIoT system in a closed-loop fashion.

1) *The “human-machine-object” View:* The interaction between the digital twin, the smart IIoT system, and the manager is illustrated by the big picture of “human-machine-object” in Fig. 2. Humans interact with the digital twin through the human-machine interface (HMI) for visualization, intelligence, and data analysis. Humans provide human intelligence based on the global information collected by the digital twin. The digital twin consists of four key components such as virtual representations, real data, simulated 3C processes, and machine learning algorithms that generate machine intelligence. Virtual representations at different granularities use different technologies such as graphics, cloud points, and others. The digital twin interacts with the smart IIoT system through digital-physical twinning. The digital twin executes resource allocation and management commands based on real-time data

collected from the system objects. The system object contains interacting network and physical components, such as cyber 3C components and physical devices.

2) *Digital Transformation:* An ideal fine-grained digital twin creates not only digital representations of physical components but also cyber 3C processes. The digital twin needs to create virtual representations for different application domains such as digital sensors and actuators for industry, digital transportation infrastructure and vehicles for transportation, and so on. The digital twin should also take into account the 3C process of intelligent IIoT systems. For example, the digital twin should simulate the network topology, data transmission tasks, channel models, and scheduling algorithms to twin the communication processes between industrial sensors, actuators, and controllers. The digital twin implements the digital control process by reproducing the control object, logic, target, system state, and environment. The digital twin reproduces scheduling algorithms, resource allocation schemes, etc. to digitize the computing process. Digital twins require real data from intelligent IIoT systems, which can be used for the temporal and spatial domain analysis [14], [55].

3) *The Closed-loop Synchronization:* The synchronization between the digital twin and the intelligent IIoT system follows a closed-loop structure, which is essential to ensure the timeliness and accuracy of the digital twin [56]–[58]. The smart IIoT system transmits data and models to the digital twin. The digital twin performs state evaluation and generates control commands for the smart IIoT system [59]. The control commands are parameter tuning and control actions for the 3C subsystem to improve control performance. The closed-loop structure ensures situational awareness, fast response, and high-fidelity twinning of various smart IIoT systems.

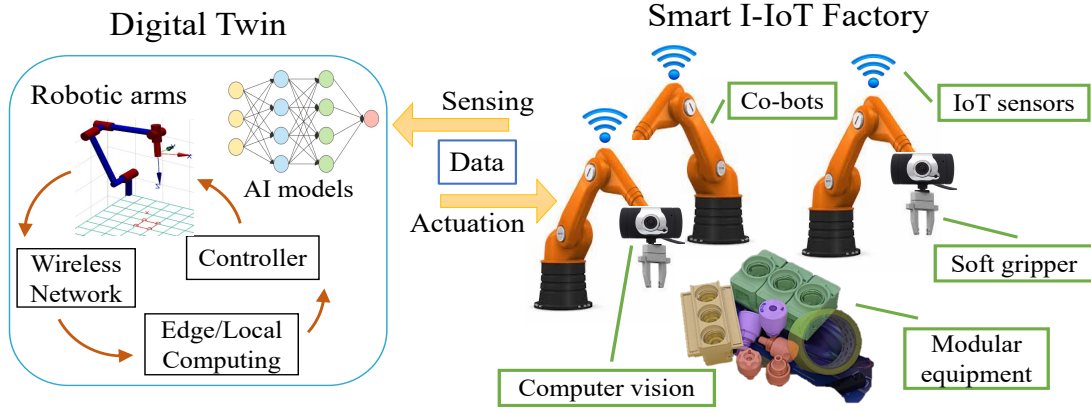


Fig. 3. Architecture of DT-IIoT Manufacturing Factories

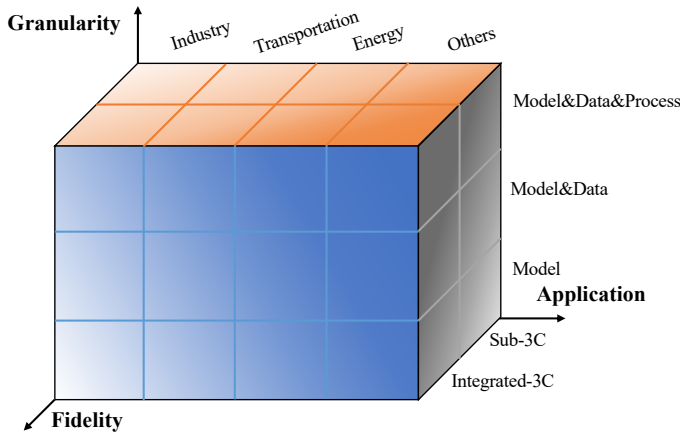


Fig. 4. The Digital Twin Three-dimensional Research Space

B. Real-World Applications of DT-IIoT

We present a three-dimensional taxonomy of digital twin research and digital twin applications.

1) *A Three-dimensional Taxonomy for Digital Twin Research*: We propose a three-dimensional research space in Fig. 4 to investigate existing digital twin applications. The taxonomy includes three dimensions such as digital twin application domain (X-axis), digital twin fidelity (Y-axis), and digital twin granularity (Z-axis).

- For the X-axis, existing digital twin application areas can be identified into four categories, including industry, transportation, energy, and others (e.g. healthcare).
- For the Y-axis, the digital twin applications can be allocated into two cases, such as the sub-3C and integrated-3C. The sub-3C indicates that the digital twin simulates the individuals of 3C processes, such as the digital twin for the software-defined vehicular network (i.e., communication) in [60], the digital twin for product management and control (i.e., control) in [61], [62], and a digital twin for multi-level computing of logistics synchronization system (i.e., computation) in [63]. The integrated-3C, which integrates the 3C subsystems, indicates a higher level of digital twin fidelity, such as a digital twin for remote control over the mobile network (i.e., integrated

communication and control) [64], mobile edge computing system with 5G (i.e., integrated computation and communication) [65].

- For the Z-axis, the digital twin applications can be classified into three granularity levels for the digital replications, such as model, data&model, and data&model&process. The coarse-grained digital twin uses simulation data for the system models and generates simulation results [7]. The data&model case uses the real data for the physical process, computing platform, and communication network models [66]. Note that simulation data is generated by following certain statistical distributions and assumptions, whereas real data is collected from real-world smart IIoT systems. For the data&model&process case, the fine-grained digital twin collects real data, system models, and intercorrelated 3C processes. The fine-grained digital twin aggregates 3C processes and considers their intercorrelations.

2) *Digital Twin of IIoT Factories*: One of the real-world applications of IIoT is the manufacturing factory, which empowers large-scale connectivity of IoT devices, such as sensors, actuators, and controllers. The DT-IIoT of the manufacturing factory creates a fine-grained digital replica of physical systems with real models and data. In addition, it can be integrated with other novel information technologies such as AI models, edge computing, operational technologies (e.g., advanced controllers), and communication technologies (e.g., 6G wireless networks) [67], [68]. Moreover, real-time interaction between the digital twin and the physical system is a key function to achieve physical state updates and command delivery as closed-loop sensing and execution.

For example, the image-based visual servoing (IBVS) system controls the linear velocity and rotational velocity of the end-effectors based on the camera inputs. The end-effectors can be soft grippers with camera sensors that can identify grip status and objects, and conduct gripping. Besides, multiple robot arms can be interconnected as co-bots to conduct complex tasks, such as welding, assembling, and others.

Fig. 3 shows the system architecture of the DT-IIoT manufacturing factories. The left side describes the digital twin modeling, which consists of the physical model of industrial

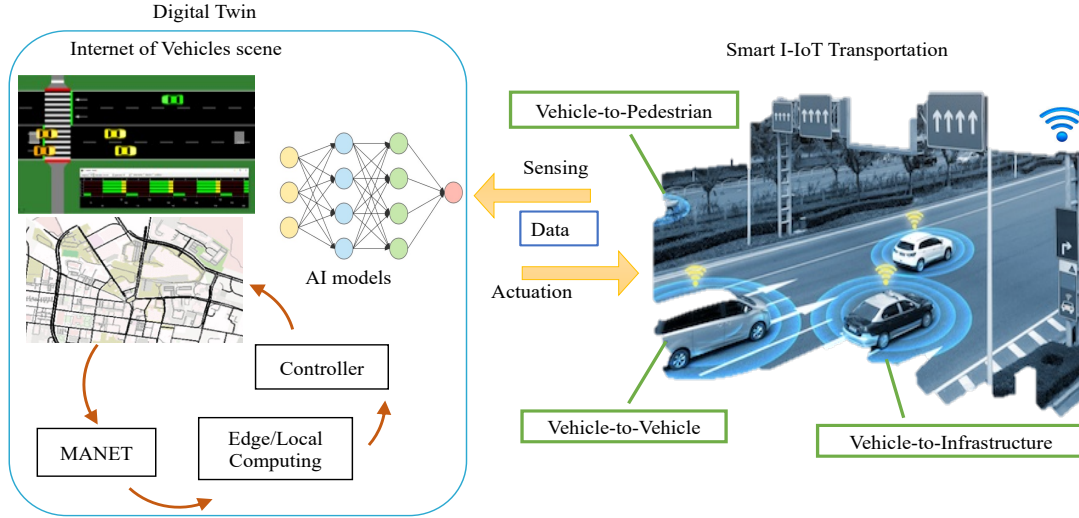


Fig. 5. Architecture of DT-IIoT Transportation Systems

robotic arms, wireless network, edge/local computing, and controller in a closed-loop fashion [69], [70]. Besides, the AI models, such as convolutional neural networks, reinforcement learning, and others are integrated for conducting prediction and automation tasks. The right side shows a scenario of the IIoT factory floor plant, which consists of sensors, actuators, and controllers, such as cameras, motors, robotic arms, and others. Moreover, the data is continuously transmitted between the left and right sides to close the loop of sensing and actuation.

3) *Digital Twin of IIoT Transportations*: Fig. 5 shows the architecture of the DT-IIoT transportation systems. The left side represents the digital twin model that simulates the real IIoT transportation systems, consisting of scenario generation, vehicle mobility modeling, network simulation, system management, and visualization modules. Thus, the digital twin model can be updated and adapted to diverse transportation scenarios by updating vehicle mobility, network protocols, communication environments, and so on. The transportation digital twin realistically reflects the interactions between physical objects and the digital replica through bi-directional real-time synchronizations. Moreover, real data is collected to support machine learning-driven applications in transportation scenarios such as the traffic flow prediction in the vehicle-to-vehicle, vehicle-to-infrastructure, and vehicle-to-pedestrian combinatorial scenarios.

C. Digital Twin Architecture and Models

This section includes the digital twin architecture and models, such as the layering model and development lifecycle model.

1) *Digital Twin Layering Model*: Based on the existing research, many reference models describe the digital twin, such as the layering model (i.e., digital twin, communication, and physical object) in [71] and the layering model (i.e., virtual factors, digital twin data, and physical factors) in [10]. However, those existing reference models have not considered the 3C subsystems to be the virtual infrastructure, which makes

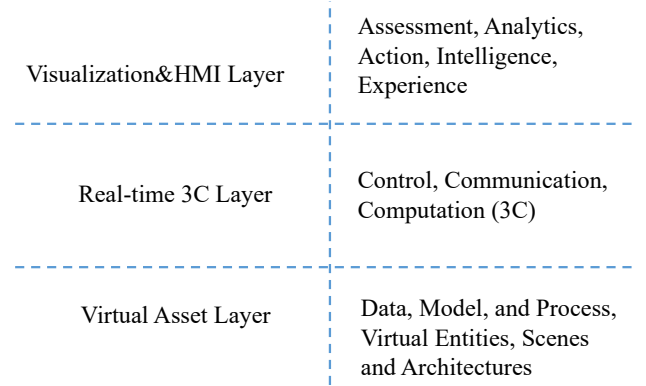


Fig. 6. The Layering Architecture of Digital Twin

the existing reference models inaccurate, coarse-grained, and non-operationalizable.

We propose the digital twin layering model in Fig. 6 that contains three conceptual layers, such as the virtual asset, real-time 3C, and visualization & HMI layers. The virtual asset layer includes the virtual assets of smart IIoT systems, including data, models, processes, virtual entities, scenes, and architectures that are the fundamental components of a digital twin. The real-time 3C layer consists of the virtual 3C processes that simulate the system processes. The visualization & HMI layer provide an interface to enable assessment, analytics, action, intelligence, and experience for the digital twin. Note that our layering model includes the 3C processes that are essential but have not been considered by others.

2) *Digital Twin Development Lifecycle Model*: The general application lifecycle follows the five-step processes, such as design, implementation, testing, monitoring, and maintenance [7], [10]. With the high-fidelity digital twin that integrates virtual representations, data, and 3C processes, advanced AI technologies can be embedded to enable prediction and automation capabilities. Therefore, we design the intelligent and secure digital twin application lifecycle, which

TABLE III. Review on Existing Research at Digital Twin Application Lifecycle Phases

Digital Twin Application Domains	AI-enabled Design	3C-based Implementation	High-fidelity Testing	Security&Operation Situation Awareness	Predictive Maintenance
Industry	[34], [72]–[81]	[61], [66], [75], [81]–[87]	[88], [89]	[33], [90]–[93]	[94], [95]
Transportation	[25], [96]	[97]	[98]	[99], [100]	[101], [102]
Energy	[103]	[104]	[105]	[106], [107]	[108]
Others	[4], [109]	[41], [110]–[113]	[64], [65], [114]	[115]	[116]

includes the AI-enabled design, 3C-based implementation, high-fidelity testing, security & operation situation awareness, and predictive maintenance, as observed in Fig. 7.

We review the existing research papers on the different stages of the intelligent and secure digital twin application lifecycle in Table III. Table III mainly allocates the digital twin of industry, transportation, energy, and other domains.

Taking industrial digital twin as an example, Schleich et al. in [72] proposed a comprehensive reference model for digital twin-based product design and manufacturing. This reference model comprehensively provides all characteristics of a physical twin at the conceptual level in the product life cycle. Different operations can then be performed on this reference model, such as combination, decomposition, transformation and evaluation. The behavior and environment of the physical twin can be captured and presented at the conceptual level. In addition, Guo et al. in [73] proposed a modular approach for flexible and time-saving factory design based on digital twins. The modular approach is a parameterization of separated modules with similar functions that are reusable and easy to build.

In specific, the AI-enabled design indicates leveraging AI algorithms to capacitate the prediction or automation capabilities of the digital twin in the design phase. There are research works, such as [74], [81], and others that can be allocated to the AI-enabled design phase.

For the implementation, the 3C-based implementation shows a fine-grained system twin that takes into account the indispensable physical communication, computation, and control subsystems. In the 3C-based implementation phase, Park et al. at [75] implemented a digital twin of the smart factory to support monitoring, tracking, and decision making functions. The digital twin was designed as a virtual factory model for information, such as configuration of manufacturing elements, synchronization information and functional units in the physical domain. Other related work such as [61], [87] can be assigned to the implementation of a 3C-based industrial digital twin.

For testing, a wide range of high-fidelity experiments with different scenarios should be conducted to validate the functionality and performance of digital twins. High-fidelity testing demonstrates the validation of digital twin models before their widespread adoption. Söderberg et al. in [88] proposes digital twin simulation for product geometric quality control in the design, pre-production and production phases. The digital twin includes geometric representation of the part, kinematic relationships, and finite element analysis to simulate the propagation of changes during assembly. In addition,

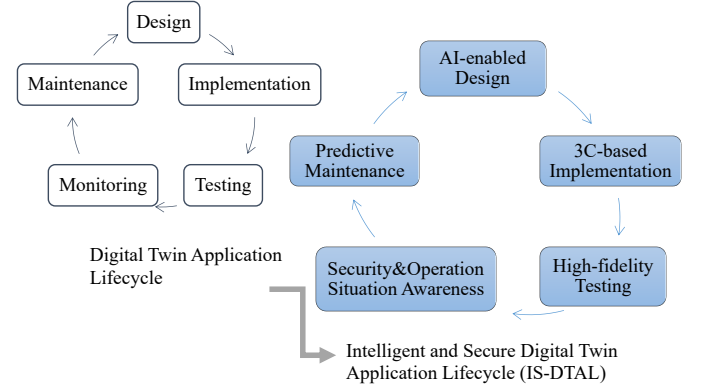


Fig. 7. The Intelligent and Secure Digital Twin Application Lifecycle

product inspection data is used in conjunction with the digital twin to detect and control production quality. In this paper, the digital twin is applied to the inspection preparation and process control of production, which can be assigned during the high-fidelity testing phase.

In terms of monitoring, the digital twin enables operational context awareness and safety awareness, where safety assurance algorithms are implemented. For safety & operational context awareness, Leng et al. in [90] investigated digital twin-driven manufacturing systems with monitoring and parallel control capabilities. Programmable logic controller (PLC) models were employed to link and embed objects into process control, industrial Internet protocols, heterogeneous device internal APIs, and data sets. bindings and mappings between PLC I/O addresses and network models enabled real-time data synchronization. Other related research work such as [91], [92] can be assigned in the safety/operational situation awareness phase.

Finally, for maintenance, predictive maintenance is enabled through AI algorithms with the massive volume of data in the digital twin. To enable the predictive maintenance, Aivaliotis et al. in [94] proposed the digital twin-based remaining useful life prediction for manufacturing machinery, similar to the digital twin-driven predictive maintenance of CNC machine tool, proposed by Luo et al. in [95].

As seen in Table III, there are more research papers focused on the design, implementation, and testing phases than on the situation awareness and predictive maintenance phases. The main reason is that predictive maintenance requires the accuracy of machine learning predictive models, which relies heavily on models and data. However, current digital twin

models are coarse-grained and industrial data are not sufficiently accessible. In addition, situational awareness requires a near-instant and high-throughput communication network for synchronization between the physical system and the digital twin, such as 6G, which is not currently available. There are significantly more applications in industry than in other application areas, such as transportation and energy. This is due to the fourth industrial revolution, also known as Industry 4.0, where the integration of IoT technologies into industry has been a trending research topic for many years. However, we can foresee that the digital twin can further accelerate the development of industry, transportation, and energy by integrating intelligent decision making and safe operation solutions [117]–[119].

D. Lessons Learned in DT-IIoT Overview

We summarize the key digital twin features from an economic, intelligence, and security perspective, as shown in the figure 8.

From an economic perspective, digital twins enable low-cost and high-fidelity manipulation for smart IIoT systems for a wide range of experiments. This reduces the cost of developing coarse-grained and asynchronous simulation, emulation, and testbed tools. The digital twin enables efficient and reliable management through resource scheduling algorithms to reduce operational costs. The digital twin also empowers rapid response and recovery, which greatly reduces financial losses following a cyber attack or emergency scenario.

From an intelligence perspective, the digital twin ensures adaptive resource allocation through AI algorithms with the massive amount of data in the digital twin. The digital twin simultaneously interacts with real-world smart IIoT systems to achieve self-awareness and evolution. In addition, the digital twin can be maintained predictively with AI algorithms. DT-IIoT has three security features such as risk assessment, security monitoring, remote diagnosis, and response. These features ensure the safety and security of the digital twin's operation.

To unleash the full power of the digital twin, we have summarized two requirements that must be urgently addressed, such as

- *Near-instantaneous communication network for digital twin synchronization:*

Synchronization between the digital twin and the intelligent IIoT system requires a high-throughput and near-instantaneous communication network to collect data from the system operation and transmit control signals and commands back to the intelligent IIoT system. System operation data is massive, including system parameters, operation states, models of communication networks, computing platforms, control loops, and physical processes in both the network and physical domains. However, current communication networks, such as 4G and 5G, cannot meet the performance requirements of digital twin synchronization, which requires research on 6G.

- *Efficient resource allocation and scheduling solutions:* AI algorithms, such as predictive models based on deep

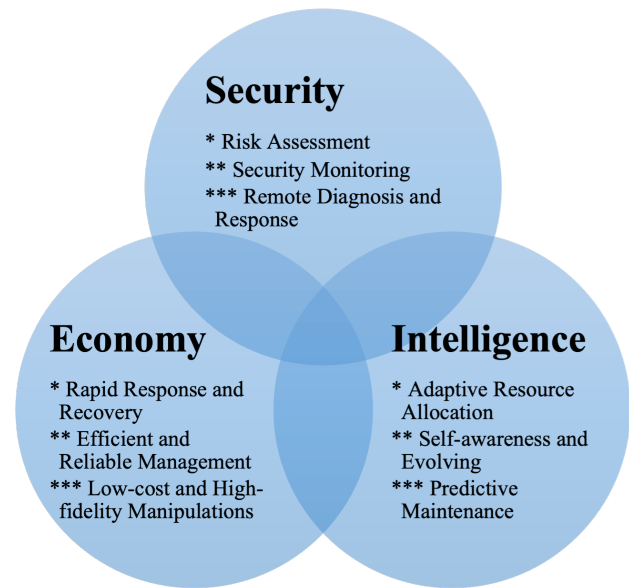


Fig. 8. The Key Digital Twin Features

neural networks, require a large number of computational resources for training. A centralized cloud computing architecture has sufficient computational resources, but it also exposes inherent poor latency performance. Therefore, hybrid cloud-edge computing architectures leverage computational resources at the edge to reduce communication latency. In addition, the problem of joint resource management and scheduling becomes increasingly complex considering highly integrated control, communication, and computation.

To meet the above requirements, AI algorithms in digital twins have great potential to enable intelligent analysis and decision-making. AI algorithms have been used for the prediction, classification, and automation of many consumer applications, such as natural language processing, computer vision, etc., which pave the way for the adoption of AI algorithms in digital twins. Therefore, we investigate preliminary work on intelligent solutions for digital twins in Section III.

III. AI EMPOWERED INTELLIGENT DT-IIoT

AI paradigms, such as neural networks, deep learning, and reinforcement learning are valuable data-driven techniques that solve prediction, classification, and automation problems. Digital twin interconnects the real-world domain through data exchanges, where the data becomes naturally viable to the AI algorithms. In this section, we investigate the research efforts that integrate AI and the digital twin.

A. A Three-dimensional Taxonomy of Intelligent DT-IIoT

We propose a three-dimensional research space in Fig. 9 for the investigation. Fig. 9 is composed of three orthogonal

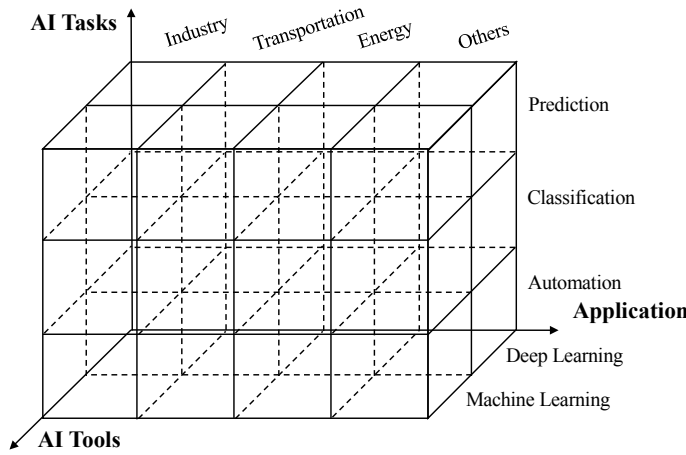


Fig. 9. The Three-dimensional Research Space of Intelligent DT-IIoT

dimensions (i.e., applications, AI tools, and AI tasks). The application areas of digital twin include industry, transportation, energy, and other domains. AI tools include machine learning and deep learning. The AI tasks are prediction, classification, and automation. Each cube in Fig. 9 represents the intersection of the three orthogonal dimensions and assigns the existing research efforts. We use Table III-D to compare the research work that can be assigned to the cubes in Fig. 9 with the problems, methods, and highlights.

B. Prediction Research of Intelligent DT-IIoT

The problem of prediction in digital twins is caused by the need for prediction in smart IIoT systems, such as predictive maintenance [120], risk prediction [121], quality prediction [122], and so on. Enabling predictive capabilities in an intelligent IIoT system can greatly improve the productivity, efficiency, and safety of physical processes. For example, predictive maintenance can predict the optimal maintenance time, thus balancing maintenance costs and equipment life and improving operational efficiency. Other predictive capabilities such as risk prediction and quality prediction have also been invented to enable safe and efficient operations. The digital twin collects models and data from real work environments and performs predictive tasks.

For industry systems, Lermer *et al.* in [123] proposed a data-driven digital twin artificial neural network and the model creation method of industrial production processes with the help of expert fuzzy rules. For the predictive maintenance case, the digital twin model incorporates the static and dynamic properties and the historical production data of the machine to obtain higher accuracy. This work can be allocated to cube $\langle X_1, Y_2, Z_3 \rangle$ in Fig. 9. Besides, Anis *et al.* in [124] proposed a digital twin-based long-term short-term memory encoder-decoder (LSTM-ED) architecture to solve the problem of estimating the machine's remaining service life. This work can be allocated to cube $\langle X_1, Y_1, Z_3 \rangle$.

C. Classification Research of Intelligent DT-IIoT

Classification problems range from anomaly detection to faulty product detection in the cyber and physical domains. Anomaly detection is an important application of classification algorithms for industrial systems. Cyber attacks against the normal operation of an industrial system may be initiated by an adversary during the system's operation, which may lead to data leakage, unsafe operations, or even security issues. Data-driven classification algorithms, such as neural networks, have been used in digital twins for anomaly detection. Digital twins collect data and describe network and physical activities in the network and physical domains and have been used to solve the anomaly detection problem [125] and the faulty node detection problem [126].

Specifically, Castellani *et al.* in [125] proposed a Siamese autoencoder architecture to conduct anomaly detection of the digital twin. The weakly supervised learning model is trained with generated training datasets from the digital twin model and a small group of anomalous datasets from the physical machine. The digital twin system outperforms the anomaly detection models with real-world datasets. Similarly, Darvishi *et al.* in [126] proposed a neural network-based architecture for faulty sensor detection, isolation, and accommodation in the industrial digital twin. Zhang *et al.* in [127] studied the stacked autoencoder and artificial neural networks based on industrial product quality monitoring in the digital twin. Those works can be allocated to cube $\langle X_1, Y_2, Z_2 \rangle$. Moreover, Xu *et al.* in [93] trained a deep neural network in the digital twin model and transferred it to the physical domain for product fault diagnosis. This work can be allocated to cube $\langle X_1, Y_1, Z_2 \rangle$.

D. Automation Research of Intelligent DT-IIoT

Automation of industrial applications requires tasks to be performed without human intervention, such as monitoring, control, and management. For example, the configuration of production systems, including hardware and software configuration of controllers and robots can be performed by digital twin [128], [129]. Automatic reconfiguration can significantly improve the adaptability of industrial systems by changing configuration parameters according to dynamic operating environments [130]. Reinforcement learning algorithms are a key approach to enhance the automation capabilities of industrial digital twins [131].

Besides, Müller-Zhang *et al.* in [131] proposed the dynamic process planning of the digital twin in industrial production systems based on reinforcement learning (i.e., Q-learning). The digital twin is treated as a virtual production environment, which contains product attributes and simulation models of resources. The reinforcement learning agent interacts with the digital twin model to derive near-optimal process planning in the dynamic production environment. Cronrath *et al.* in [132] proposed a contextual bandit-based learning approach to enhance the manufacturing digital twin. Bansal *et al.* in [133] studied the ant colony optimization-based industrial robot path planning in the digital twin. Those works can be allocated to cube $\langle X_1, Y_2, Z_1 \rangle$. Furthermore, Sun *et al.* [39] proposed a dynamic global aggregation of federated learning in the IIoT

digital twin based on deep reinforcement learning. This work can be allocated to cube $\langle X_1, Y_1, Z_1 \rangle$.

E. AI Empowered DT-IIoT for Other Domains

Some studies combine AI and digital twin to a wider range of application scenarios such as transportation [60], [134], energy [140], [141], and other domains [31], [144].

1) *Intelligent DT-IIoT of Transportation*: The digital twin of a vehicle system is a virtual system that images the activities of vehicles, traffic infrastructure, and information flows where they can be managed holistically. Based on the data collected in the traffic digital twin, predictive management, such as traffic forecasting and route management, can be performed using artificial intelligence algorithms to improve travel efficiency [147]–[149].

Work [60], [134] focusing on vehicle system prediction can be assigned to the cube $\langle X_2, Y_2, Z_3 \rangle$. Ezhilarasu et al. proposed a digital twin-based integrated vehicle health management framework for assessing and predicting the health of complex systems in [134]. Zhao et al. proposed digital twin-based traffic prediction for vehicle networks in [60]. The integration of smart DT-IIoT builds a virtual intelligent cyberspace for software-defined vehicular networks. The proposed digital twin model allows iterative updates of network schemes such as routing and network services.

Besides, Lei et al. in [135] proposed an intelligent cooperation framework for the digital twin of unmanned aerial vehicle (UAV) swarm. Uzun et al. in [136] proposed a deep learning-based fuel burn estimate for the aircraft digital twin. Work [135], [136] can be allocated to cube $\langle X_2, Y_1, Z_3 \rangle$. Mihai et al. in [137] uses support vector machine (SVM) to conduct pedestrian detection can be allocated in $\langle X_2, Y_2, Z_2 \rangle$. Saifutdinov et al. in [138] proposed digital twin-assisted decision-making for airport traffic control can be allocated to cube $\langle X_2, Y_2, Z_1 \rangle$. The digital twin is designed based on the ground traffic scenario simulation model, which continuously generates data flows of airport traffic and trains the machine learning-based airport traffic control systems. Xu et al. in [139] proposed a deep reinforcement learning-based service offloading method for the digital twin of the edge internet of vehicles that can be allocated to cube $\langle X_2, Y_1, Z_1 \rangle$. The digital twin-empowered IoV is developed to collect the state information of vehicles and infrastructures to support the data-driven computing resource management algorithms.

2) *Intelligent DT-IIoT of Energy*: A smart grid is a valuable application that leverages information and communication technologies for electricity management, such as energy trading [150], [151]. The integration of digital twin and AI can help the smart grid in achieving intelligent optimization and automated energy management [152]

Few works address the problems in the energy application domains using AI and digital twin [140], [141]. Snijders et al. in [140] proposed a temporal convolutional neural network (TCN) based prediction on battery responsiveness in digital twin for energy systems. Xie et al. in [141] proposed a deep learning-based demand prediction for electrical digital twin. Those works can be allocated to cube $\langle X_3, Y_2, Z_3 \rangle$.

3) *Other Intelligent DT-IIoT Applications*: Some works combine the digital twin and AI for other communication systems, such as 5G, 6G, and so on. Wang et al. in [142] proposed the graph neural networks based on end-to-end latency prediction for the digital twin of a 5G communication network. This work can be allocated to cube $\langle X_4, Y_2, Z_3 \rangle$. Martinez-Velazquez et al. in [49] proposed deep learning-based cardiography analysis and classification in the digital twin of healthcare. El Marai et al. in [143] proposed K-means clustering based cluster creations for digital twin for the internet of battlefield things. Those works can be allocated to cube $\langle X_4, Y_2, Z_2 \rangle$.

For automation, Szabó et al. in [144] proposed reinforcement learning (actor-critic) based radio access control for digital twin of 5G communication network. Sun et al. in [42] proposed a Stackelberg game-based incentive mechanism for federated learning in the digital twin of air-ground network. Kulkarni et al. in [153] proposed a reinforcement learning-based adaptive enterprise digital twin. Those works can be allocated to cube $\langle X_4, Y_2, Z_1 \rangle$.

Besides, Lu et al. in [31] proposed a multi-agent deep reinforcement learning-based trust federated learning framework for edge association in the digital twin of 6G communication network. Dong et al. in [65] proposed a deep learning framework for user association 5G mobile edge computing. Sun et al. in [41] proposed a deep reinforcement learning based task offloading scheme for digital twin of 6G edge network. Dai et al. in [145] proposed a deep reinforcement learning-based computation offloading for the digital twin of communication network. Cui et al. in [146] proposed a deep reinforcement learning based adaptive configuration for digital twin of optical transceiver. Those works can be allocated to cube $\langle X_4, Y_1, Z_1 \rangle$.

F. The State-of-the-art AI Solutions for Intelligent DT-IIoT

We now investigate state-of-the-art AI algorithms, such as transfer learning, federated learning, and so on that solve data scarcity and efficiency problems during model training.

1) *Transfer Learning-based Efficient Digital Twin Learning Model*: The data-driven approach plays a vital role in solving prediction, classification, and automation problems of the digital twin, such as the remaining useful life prediction, fault diagnosis, automated reconfiguration, etc. To train a reliable model, we need a sufficient amount of training data and training time. But, in practice, the datasets are usually insufficient in many application domains, such as industrial, transportation and energy, etc [154]. Besides, the time-consuming training process hinders the adaptability and decision-making of the digital twin.

The transfer learning technique leverages a well-trained learning model and transfers it to a new model through the tuning process. The new model can be adapted to solve a related problem with little training data and training time [155]. The well-trained deep learning model works as a feature extractor that can be reused to train new prediction or classification models. The prerequisites are that the data used to train the original learning model and the data used to train the

TABLE IV. Review on Research of Intelligent DT-IIoT with Three-dimensional Taxonomy in Fig. 9

Cube	Existing Work	Problem	Approach	Highlight
$\langle X_1, Y_2, Z_3 \rangle$	Lerner et al. [123]	predictive maintenance for manufacturing	artificial neural networks	leveraged the fuzzy rules from experts to assist model creation
$\langle X_1, Y_1, Z_3 \rangle$	Anis et al. [124]	remaining useful life estimation	long-short term memory (LSTM) recurrent neural network	created a binary classification model for the health state estimation
$\langle X_1, Y_2, Z_2 \rangle$	Castellani et al. [125]	anomaly detection	clustering algorithm and Siamese autoencoder architecture	created a weakly-supervised classification model for infrequent anomalous data samples
-	Darvishi et al. [126]	anomaly detection	neural networks based estimation and classification architecture	created the faulty sensor detection, isolation, and accommodation architecture
-	Zhang et al. [127]	industrial product quality monitoring	stacked autoencoder and artificial neural networks	created the parameters-update rule to improve model accuracy
$\langle X_1, Y_1, Z_2 \rangle$	Xu et al. [93]	product fault diagnosis	deep transfer learning	transferred the diagnosis model from virtual space to physical space
$\langle X_1, Y_2, Z_1 \rangle$	Müller-Zhang et al. [131]	dynamic process planning	Q-learning	applied the Q-learning algorithm to derive process plans
-	Cronrath et al. [132]	improving manufacturing digital twin	contextual bandit and deep learning	enhanced the control policy of digital twins in continuous domains
-	Bansal et al. [133]	industrial robot path planning	ant colony optimization	conducted robot path planning in digital twin
$\langle X_1, Y_1, Z_1 \rangle$	Sun et al. [39]	learning model efficiency	deep reinforcement learning and federated learning	proposed the dynamic global aggregation of federated learning with digital twin
$\langle X_2, Y_2, Z_3 \rangle$	Ezhilarasu et al. [134]	integrated vehicle health management	digital twin based integrated vehicle health management framework	investigated the health assessment and prediction of complex systems with digital twin
-	Zhao et al. [60]	traffic prediction for vehicular networks	intelligent digital twin-based software-defined vehicular networks (IDT-SDVN)	investigated the IDT-SDVN
$\langle X_2, Y_1, Z_3 \rangle$	Lei et al. [135]	intelligent cooperation of unmanned aerial vehicle (UAV) swarm	deep neural network	proposed a digital twin based UAV swarm control framework
-	Uzun et al. [136]	aircraft fuel burn estimation	deep neural network	applied digital twin model to actual filed flight plans
$\langle X_2, Y_2, Z_2 \rangle$	Mihai et al. [137]	pedestrian detection	support vector machine (SVM)	used lite LiDAR technology
$\langle X_2, Y_1, Z_2 \rangle$	-	-	-	-
$\langle X_2, Y_2, Z_1 \rangle$	Saifutdinov et al. [138]	airport traffic control	neural network	studied the digital twin assisted decision making
$\langle X_2, Y_1, Z_1 \rangle$	Xu et al. [139]	service offloading in the internet of vehicles (IoV)	deep Q-network (DQN)	studied the integration of DQN and digital twin of edge IoV
$\langle X_3, Y_2, Z_3 \rangle$	Snijders et al. [140]	battery responsiveness prediction	temporal convolutional neural network (TCN)	studied the response prediction in digital twin for energy system
-	Xie et al. [141]	energy demand prediction	deep neural network	proposed a digital twin forecasting framework with RNN and neural ordinary differential equation (ODE)
$\langle X_3, Z_1, Z_3 \rangle$	-	-	-	-
$\langle X_3, Y_2, Z_2 \rangle$	-	-	-	-
$\langle X_3, Y_1, Z_2 \rangle$	-	-	-	-
$\langle X_3, Y_2, Z_1 \rangle$	-	-	-	-
$\langle X_3, Y_1, Z_1 \rangle$	-	-	-	-
$\langle X_4, Y_2, Z_3 \rangle$	Wang et al. [142]	latency prediction for 5G	graph neural networks	studied the end-to-end latency prediction in digital twin for network slicing management
$\langle X_4, Y_1, Z_3 \rangle$	-	-	-	-
$\langle X_4, Y_2, Z_2 \rangle$	Martinez-Velazquez et al. [49]	cardiography analysis and classification	deep neural networks	proposed an architecture of a digital twin for healthcare
-	El Marai et al. [143]	clustering in the internet of battlefield things	K-means	investigated the roads infrastructure digital twin
$\langle X_4, Y_1, Z_2 \rangle$	-	-	-	-
$\langle X_4, Y_2, Z_1 \rangle$	Szabó et al. [144]	radio access control in 5G	actor-critic	transferred the learned policies in digital twin to real-world environment
-	Sun et al. [42]	learning model efficiency	Stackelberg game based incentive mechanism	designed the dynamic digital twin of air-ground network for federated learning tasks
$\langle X_4, Y_1, Z_1 \rangle$	Lu et al. [31]	edge association	multi-agent deep reinforcement learning	optimized the joint communication scheduling and edge association in the digital twin of 6G communication network
-	Dong et al. [65]	user association	deep neural networks	trained the deep learning algorithm in digital twin to optimize the user association, resource allocation, and offloading probabilities jointly
-	Sun et al. [41]	task offloading	actor-critic deep reinforcement learning	proposed the digital twin edge network
-	Dai et al. [145]	computation offloading	asynchronous actor-critic deep reinforcement learning	proposed the digital twin network with virtual models of IIoT entities for computation offloading
-	Cui et al. [146]	adaptive configuration	deep reinforcement learning	proposed the adaptive digital twin based programmable optical transceiver

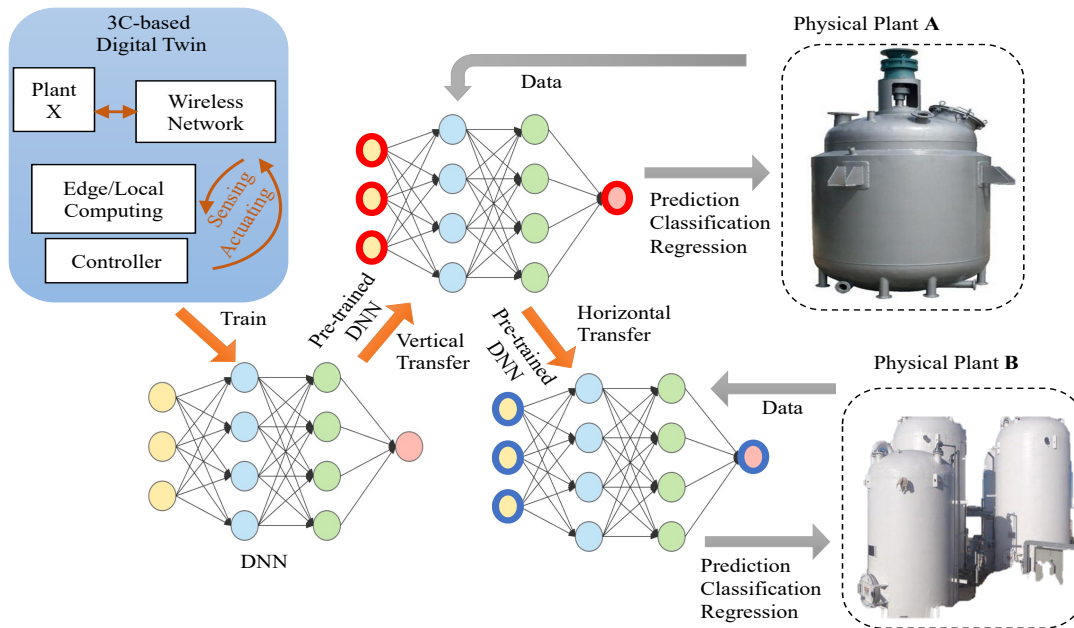


Fig. 10. Vertical and Horizontal Deep Transfer Learning Assisted Digital Twin

transferred model must share some common features and the original problem and the new problem must be related [154].

We use Fig. 10 to present the vertical and horizontal ways of leveraging transfer learning to reduce the training time of deep learning models for digital twin.

- **Vertical Transfer Learning:** Vertical transfer learning indicates transferring the learning model from virtual space (i.e., digital twin) to physical space (i.e., physical plant). Specifically, the 3C-based digital twin, which is consistent with the smart IIoT systems in physical space can generate a sufficient amount of data without interfering with the operations. Then, the deep learning model is trained and tested by the collected digital twin data. Finally, transfer learning transfers the well-trained model to physical space (physical plant A) through tuning. Similarly, Xu *et al.* in [93] proposed digital-twin-assisted fault diagnosis with a deep transfer learning technique. In their approach, they leveraged the digital twin data to train the DNN diagnosis model and transferred it to the physical space.
- **Horizontal Transfer Learning:** Horizontal transfer learning is transferring the learning model from physical space A to physical space B directly. Specifically, the learning model is trained to solve learning problems of physical plant A. Then, this well-trained model can be transferred to solve the learning problems of physical plant B by following the above-mentioned processes. Recall that the learning problems of physical plants A and B must be related. Liu *et al.* in [154] adopted transfer learning to solve the industrial image classification with limited datasets problem for industrial internet-of-things.

2) **Federated Learning-based Secure Digital Twin Learning Model:** To conduct efficient model training, we usually collect and pre-process the training data in a centralized and unified

fashion. The digital twin of a smart IIoT system is often treated as the control and data center, which collects data from the physical devices and carries out the model training [156]–[158]. With the fast-growing industrial, transportation, and medical smart IIoT systems, the massive volume of data to be collected and delivered to the digital twin could cause network overload problems, low efficiency, and data security risks.

Federated learning is a machine learning paradigm that trains machine learning models on multiple distributed devices collaboratively without sharing the raw data [159]–[161]. The distributed devices use their local data to train a general learning model individually. Then, the devices only upload the model parameters to the aggregator, which aggregates all trained model parameters and returns the model to distributed devices for further training. This iterative training process continues until the learning model converges. Federated learning can effectively improve communication efficiency, and learning efficiency and ensure data security. Likewise, federated learning for the digital twin can improve the digital twin model efficiency and data security by limiting the data transfer to the digital twin and reducing the data sharing among virtual space and physical subsystems.

We use Fig. 11 to demonstrate the architecture of the federated learning-assisted digital twin. The cyber domain consists of 3C subsystems, which interact with the physical devices through sensing and actuating. The digital twin interacts with the smart IIoT system through data exchange, which reflects the physical dynamics of the system. The 3C-based digital twin establishes the general learning model for diverse machine learning tasks. The general learning model is trained by the 3C subsystems with their computing, communication, and control data, following the federated learning iterative training process. In this way, the general learning model in the 3C-based digital twin can be trained without extensive

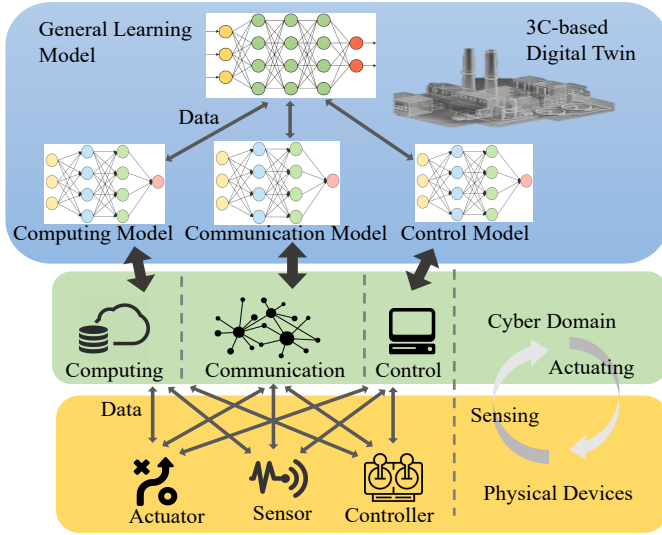


Fig. 11. Federated Learning Model for Digital Twin

data exchanges between the digital twin and the system, which ensures data security.

Lu *et al.* in [162] proposed the federated learning-based digital twin edge network for the IIoT. Regarding the resource-constrained IIoT devices, the authors developed the asynchronous model updating scheme to improve communication efficiency for federated learning. The digital twin is created to be the virtual network that the physical network states. Sun *et al.* in [39] integrated the federated learning with the digital twin of a dynamic air-ground network. The digital twin can capture the dynamics of elements in the air-ground network and carry out decision-making. The resource-constrained ground devices train the model using their data and send model parameters to a drone for aggregation. Then, the drone updates the ground devices. The authors proposed a Stackelberg game-based dynamic incentive mechanism for ground devices in responding to the dynamics of the air-ground network.

G. Lessons Learned in AI Empowered Intelligent DT-IIoT

Based on the above investigations, we draw a few conclusions as follows:

- Automation is the fundamental characteristic of the intelligent DT-IIoT, observed from a large number of research efforts on automation.
- There is a limited number of research works on prediction. This indicates that the digital twin anticipates the prediction capability for predictive tasks, which requires additional data and research efforts.
- Compared to machine learning algorithms, deep learning has not been adopted widely for intelligent tasks in the digital twin. Further investigations on the deep learning for digital twin is in-demand, which exploits the DNN's capabilities in approximating complex functions or models.
- There are more works in the industry and transportation domains, compared to the energy domain. The reason is that the high system dynamicity in the industry and

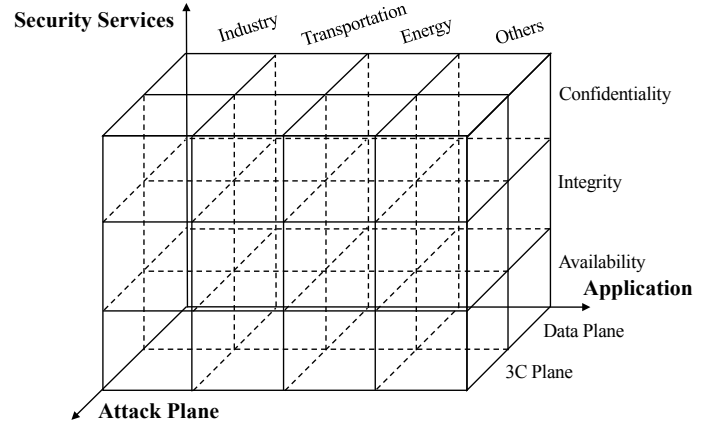


Fig. 12. The Three-dimensional Research Space of DT-IIoT Security

transportation domains requires adaptive and predictive mechanisms. However, the high system dynamicity in the energy domain also exists, such as the dynamic supply and demand relations, which raise the need for intelligent solutions.

As the digital twin becomes the intelligent center and data center. The digital twin's security, data privacy, and resilience must be guaranteed urgently. Thus, we look into the preliminary works on the security solutions of the digital twin in Section IV.

IV. DT-IIoT SECURITY

The digital twin is the intelligent center and the critical virtual asset, which highlights the urgency of safety and security assurance. The massive interactions between the virtual domain, cyber domain, and physical domain bring extendable vulnerabilities making the safety and security of the digital twin challenging. In this section, we investigate the research efforts on the security of the AI-enabled digital twin.

A. A Three-dimensional Taxonomy of Secure DT-IIoT

We propose a three-dimensional research space in Fig. 12 to study DT-IIoT security. X-axis is the digital twin application. Y-axis is the attack plane, including the data plane, the 3C plane. Z-axis is the security service, including availability, integrity, and confidentiality. In addition, we use table IV-D to compare research efforts and problems, methods, and highlights that can be assigned to the research space in Fig. 12. As observed in Fig. 12, the threat space is a combination of attack planes and security services for different applications. For example, industrial digital twins are vulnerable to attacks on security services, including integrity, confidentiality, and availability.

B. DT-IIoT Data Security

The integrity protects data from modification, deletion, and injection. To disrupt the integrity, the attacker can launch a common integrity attack, the man-in-the-middle attack, by changing the content of messages in transit. Confidentiality

protects the information secrecy by ensuring only authorized access. To disrupt confidentiality, an attacker could pose as a legitimate node and dupes users to connect. Once connected, the attacker can use the malicious node to obtain the user's confidential information. The availability ensures the industrial digital twin functions are available to users. The attacker could launch the signal jamming attack to disrupt communication availability and remain undetected.

The attack plane is the other domain of the cyberattacks, such as the data plane and 3C plane. The data plane mainly collects, transports, and stores data for the industrial digital twin. The 3C plane indicates the signaling processes for control, communication, and computing, such as synchronization, security, authentication, sensing, actuating, scheduling, and so on.

Malicious users can launch attacks in a combination of an attack plane and security services, such as the integrity and data plane. For example, some legitimate users become compromised to form a mobile botnet. The botnet could overwhelm legitimate users by flooding the transportation level. This type of attack can be allocated to the attack combination of availability and data plane.

There are research efforts that studied the attack scenarios, such as the combination of security services and the data plane in the context of the industrial digital twin. For the attack combination of integrity and data plane, Akbarian *et al.* in [163] proposed the support vector machine-based false data injection attacks detection and classification in the digital twin of industrial control system. Xu *et al.* in [164] studied Chi-square detector-based state estimation attack detection in industrial digital twin. Dietz *et al.* in [165] proposed a process-based security framework for attack simulation (i.e., the man-in-the-middle attack) in the digital twin of industrial operations center. Those works can be allocated to cube $\langle X_1, Y_1, Z_2 \rangle$ in Fig. 12.

Besides, for the attack combination of confidentiality and data plane, Putz *et al.* in [166] studied the blockchain-based access control to ensure confidentiality for data sharing in industrial digital twin. Mandolla *et al.* in [167] studied the blockchain-based additive manufacturing architecture for the digital twin of the metal manufacturing process. Those works can be allocated to cube $\langle X_1, Y_1, Z_3 \rangle$.

C. DT-IIoT 3C Security

The digital twin 3C plane security studies the security issues that hinder the availability, integrity, or confidentiality of the digital twin of 3C signaling processes, such as synchronization, security, authentication, sensing, actuating, scheduling, and so on. For example, adversaries establish malicious connections without transmitting any data. Those malicious connections reduce available bandwidth and resources for legitimate users. The attack combination is the availability and control plane.

For the attack combination of availability and 3C plane, Alves *et al.* in [168] studied the implementation of the denial-of-service (DoS) and man-in-the-middle attacks in the virtual industrial control system testbed. This work can be allocated to cube $\langle X_1, Y_2, Z_1 \rangle$.

For the attack combination of integrity and 3C plane, Eckhart *et al.* in [169] proposed a security-aware framework for MITM attack detection in digital twin. Dietz *et al.* in [170] studied the security-operation modes for attack detection in industrial digital twin. Those works can be allocated to the cube $\langle X_1, Y_2, Z_2 \rangle$.

For the attack combination of confidentiality and 3C plane, Besides, Gehrmann *et al.* in [5] proposed a security framework to protect the confidentiality and integrity of the digital twin of industrial control and automation systems. This work can be allocated to cube $\langle X_1, Y_2, Z_3 \rangle$.

D. DT-IIoT Security Issues for Other Applications

We use the security space in Fig. 12 to investigate the research on DT-IIoT security of other applications, such as transportation, energy, healthcare, and so on. For transportation DT-IIoT security, Damjanovic-Behrendt in [171] proposed a method to enhance the digital twin privacy for smart car. Kanak *et al.* in [172] studied the blockchain-based collaborative processes for the digital twin of smart transportation and proposed a blockchain-based accountability architecture. Those works can be allocated to the cube $\langle X_2, Y_1, Z_3 \rangle$.

For energy DT-IIoT security, Danilczyk *et al.* in [173] proposed a digital twin security framework for the microgrid to detect cyber attacks, such as denial-of-service attacks. Atalay *et al.* in [174] proposed a digital twin security evaluation framework for risk evaluation of smart grid. Saad *et al.* in [104] studied the digital twin-based detection and mitigation of the cyber attacks (denial-of-service attacks) in microgrids. Those works can be allocated to cube $\langle X_3, Y_2, Z_1 \rangle$.

For DT-IIoT security of other applications, Lu *et al.* in [32] studied the blockchain-based federated learning with security and data privacy assurance for the digital twin of edge communication network. Maillet-Contoz *et al.* in [175] proposed an end-to-end security layer for security validation for IoT system in digital twin. Zhang *et al.* in [176] studied the deep learning-based vulnerability detection for healthcare digital twin. Hasan *et al.* in [177] studied the blockchain-based digital twin creation to resist security vulnerabilities and attacks. Dietz *et al.* in [178] proposed the distributed ledger-based digital twin data sharing for asset lifecycle management. Those works can be allocated to cube $\langle X_4, Y_1, Z_3 \rangle$.

E. The State-of-the-art Blockchain Solutions for DT-IIoT Security

1) *Blockchain Overview:* Blockchain is the state-of-the-art solution to build consensus in a large-scale peer-to-peer network. Leveraging blockchain can enable the benefits of decentralization, legitimacy, traceability, and trust to users [179], [180]. The key characteristic of Blockchain is the linked blocks, where each block contains the cryptographic hash value of the previous block, a timestamp, and transaction data. The hash value is computed by the cryptographic hashing function with the parent block as input. The timestamp is stored in blocks to record the time of block creation. The chained blocks store data in a distributed fashion and ensure the transaction data are unalterable. This is because the hash

TABLE V. Review on Research of DT-IIoT Security with Three-dimensional Taxonomy in Fig. 12

Cube	Existing Work	Problem	Approach	Highlight
$\langle X_1, Y_1, Z_1 \rangle$	-	-	-	-
$\langle X_1, Y_1, Z_2 \rangle$	Akbarian et al. [163]	false data injection attacks detection and classification	support vector machine (SVM)	proposed the Kalman filter and SVM based attack detector and classifier in the digital twin of industrial control system
-	Xu et al. [164]	state estimation attack detection	Chi-square detector	studied the optimal attack and defense strategies with game theory in the digital twin of cyber-physical system
-	Dietz et al. [165]	attack simulation (i.e., a man-in-the-middle attack)	process-based security framework	proposed the process-based security framework with security simulation in digital twin
$\langle X_1, Y_1, Z_3 \rangle$	Putz et al. [166]	access control	Blockchain	proposed the distributed ledgers based digital twin data sharing
-	Mandolla et al. [167]	creating a secure manufacturing infrastructure	Blockchain	proposed Blockchain-based additive manufacturing architecture for metal manufacturing process
$\langle X_1, Y_2, Z_1 \rangle$	Alves et al. [168]	creating control system testbeds	virtualization	created the virtual industrial control system testbed and simulated the denial-of-service (DoS) and MITM attacks
$\langle X_1, Y_2, Z_2 \rangle$	Eckhart et al. [169]	intrusion detection	security-aware framework	proposed the twinning framework to generate and execute digital twin in an industrial scenario
-	Dietz et al. [170]	attack detection	security-operation modes	investigated the potentials of a digital twin for industrial control systems security
-	Alves et al. [168]	creating control system testbeds	virtualization	implemented the DoS and MITM attacks in the virtual industrial control system testbed
$\langle X_1, Y_2, Z_3 \rangle$	Gehrmann et al. [5]	industrial system security architecture	digital twin	identified the security requirements of digital twin based security architecture for industrial control and automation systems
$\langle X_2, Y_1, Z_1 \rangle$	-	-	-	-
$\langle X_2, Y_1, Z_2 \rangle$	-	-	-	-
$\langle X_2, Y_1, Z_3 \rangle$	Damjanovic-Behrendt [171]	digital twin privacy	-	studied the digital twin-based privacy enhancement method for smart cars
-	Kanak et al. [172]	secure collaborative processes in digital twin	Blockchain	proposed the Blockchain-based model for digital twin systems
$\langle X_2, Y_2, Z_1 \rangle$	-	-	-	-
$\langle X_2, Y_2, Z_2 \rangle$	-	-	-	-
$\langle X_2, Y_2, Z_3 \rangle$	Kanak et al. [172]	accountable digital twin architecture	Blockchain	proposed the Blockchain-based model for digital twin systems
$\langle X_3, Y_1, Z_1 \rangle$	-	-	-	-
$\langle X_3, Y_1, Z_2 \rangle$	-	-	-	-
$\langle X_3, Y_1, Z_3 \rangle$	-	-	-	-
$\langle X_3, Y_2, Z_1 \rangle$	Danilczyk et al. [173]	detecting cyber attacks	security framework	proposed a digital twin framework to improve the security and resiliency of the microgrids
-	Atalay et al. [174]	smart grid risk evaluation	security evaluation framework	proposed a digital twin based security evaluation framework for smart grid
-	Saad et al. [104]	detection and mitigation of the DOS attacks	digital twin	proposed the IoT based digital twin of networked microgrids
$\langle X_3, Y_2, Z_2 \rangle$	Danilczyk et al. [173]	detecting cyber attacks	security framework	proposed a digital twin framework to improve the security and resiliency of the microgrids
-	Saad et al. [104]	detection and mitigation of the FDI attacks	digital twin	proposed the IoT based digital twin of networked microgrids
$\langle X_3, Y_2, Z_3 \rangle$	-	-	-	-
$\langle X_4, Y_1, Z_1 \rangle$	-	-	-	-
$\langle X_4, Y_1, Z_2 \rangle$	-	-	-	-
$\langle X_4, Y_1, Z_3 \rangle$	Lu et al. [32]	security and data privacy assurance for federated learning	Blockchain	integrated the digital twin edge networks (DITENs) with blockchain-empowered federated learning
-	Maillet-Contoz et al. [175]	security validation	end-to-end security layer	-
-	Zhang et al. [176]	vulnerability detection	bidirectional LSTM	addressed the vulnerability detection considering code context relationships
-	Hasan et al. [177]	resisting security vulnerabilities and attacks	Blockchain	proposed the digital twin creation processes with blockchain
-	Dietz et al. [178]	data sharing	distributed ledger	leveraged distributed ledger to secure digital twin data sharing
$\langle X_4, Y_2, Z_1 \rangle$	-	-	-	-
$\langle X_4, Y_2, Z_2 \rangle$	-	-	-	-
$\langle X_4, Y_2, Z_3 \rangle$	-	-	-	-

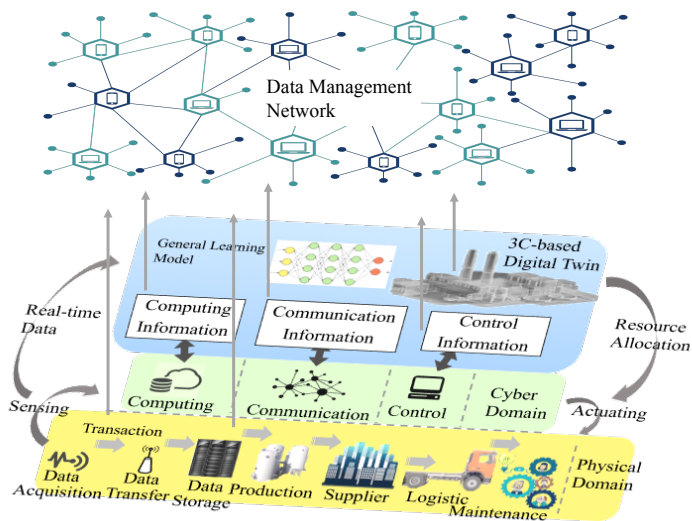


Fig. 13. The Blockchain-enabled Data Management Network for Digital Twin

function is a one-way encryption function, which turns any length input into a unique cryptographic fixed output. Any reordering and modification of the blocks will be detected easily. The distributed architecture can minimize the impact of the individual failure of nodes and improve the system's robustness [181], [182].

There are three types of blockchain structures: public, private, and consortium [183]. A cryptocurrency, such as BTC follows a public blockchain structure, where it can be accessed by anyone on the network. The private blockchain is usually constructed and managed by private organizations. The consortium blockchain follows a semi-private structure, which connects multiple private organizations and ensures a certain level of privacy. Consortium blockchain has many advantages, such as faster reaching consensus, high throughput, and privacy, compared to public and private blockchains.

The consensus algorithms play a critical role in making decisions for a group in a decentralized and trustless environment. The purpose of reaching a consensus is to trustfully validate the newly generated blocks with the presence of malicious nodes. Individuals exchange messages for multiple rounds following the pre-defined consensus sequences to reach a consensus. The well-known consensus algorithms are Proof of Work (PoW), Proof of Stake (PoS), Practical Byzantine fault-tolerant Mechanism (PBFT), and so on. The PoW is competing for the speed of the mathematical puzzle-solving problem. The first participant who solved the puzzle correctly can generate a new block in the blockchain. The puzzle-solving problem is to find a nonce that makes the hash value of the block header smaller than the specific target value. The limitations of PoW are the low efficiency, resource-consuming, and time-consuming [184].

For time-sensitive and resource-constraint applications, the excessive delay and resource-consuming in reaching consensus is the critical challenge faced by consensus algorithms. The lightweight PBFT can effectively reach a consensus with the existence of faulty nodes. In PBFT, all participants are

ordered in a sequence with one leader and others to be backup nodes. Leader multicasts the client's request to all backup nodes. The backup nodes process the request and reply to the client. The client receives at least $f + 1$ replies with the same result to be the final result. The f is the tolerable faulty node in the network and the f is smaller than the $1/3$ of total nodes [185]. Compared to PoW, the PBFT is computing-friendly, easy to implement (i.e., lightweight), and faster in reaching consensus. However, the communication overhead increases with the network size going up, which is the major limitation. Dai *et al.* in [186] concluded two types of consensus algorithms in the context of IoT, such as the probabilistic consensus, including PoW and PoS, and deterministic consensus, such as PBFT and PBFT variants. The current research on blockchain-based digital twin mainly focuses on industry [166], [167], transportation [172], and other domains [32], [187].

2) *Blockchain Assisted Digital Twin Systems*: Recall Section II, a smart IIoT system is an integrated system that synthesizes cyber components and physical entities through closed-loop sensing and actuating. The digital twin is the digital representation of the smart IIoT system, which interacts with the smart IIoT system through real-time data collection and resource allocation. The data management, including data storage and sharing of the digital assets in digital twin requires stronger data protection mechanisms to ensure the data integrity [186], [188], [189]. Blockchain enables the decentralized data management method and protects data security and privacy in data generation, sharing, and storage processes [190], [191]. We use Fig. 13 to demonstrate the architecture of the blockchain-enabled data management network for the digital twin. The blockchain is the distributed database that has the access to data across "virtual-cyber-physical" domains, which improves data management efficiency and security vastly.

Integrating blockchain into the digital twin is a trending research topic and has attracted many research efforts. Yaqoob *et al.* in [180] investigated the research articles on leveraging the blockchain into a digital twin for secure manufacturing. They summarized the key benefits as immutability, legitimacy, traceability, transparency, and so on. Besides, key challenges that are to be solved, such as the scalability issue, network stress, and privacy issue of the distributed nodes were discussed in the paper.

Towards collaborative smart manufacturing, Tao *et al.* in [189] proposed the digital twin and blockchain-enhanced industrial internet platform. Driven by the integrated data and model, the digital twin can significantly improve efficiency and accuracy in coordinating manufacturing service collaboration. The blockchain is introduced to manage the collaborations and transactions of all participants, including customers, platforms, and stakeholders, throughout the manufacturing service collaboration process. The PBFT is adopted to be the consensus algorithm. Huang *et al.* in [192] established a peer-to-peer network to construct the blockchain, which manages data throughout the product lifecycle, including data storage, sharing, access, authentication, and so on. The data is recorded as transactions and stored in the blocks, including digital twin

actions, sensor data, and so on. They conducted a case study on blockchain-based data management for the digital twin of a turbine and adopted the PoW consensus algorithm.

For other application domains, such as construction projects, Lee *et al.* in [193] developed the traceable data communication framework for supply chain management with digital twin and blockchain technologies. The digital twin for construction enables the “up-to-current” construction modeling through automatic updating. The blockchain enables efficient collaboration among all participants of a large construction project safely and trustily.

F. Lessons Learned in DT-IIoT Security

Based on the investigation of the security of the AI-enabled digital twin, we make a few conclusions as follows:

- Digital twin establishes a new virtual domain and empowers the massive data exchanges across “virtual-cyber-physical” domains, which makes its security problem challenging.
- Effective defense algorithms follow the cross-domain security assurance methodology, such as the cyber-physical integrated intrusion detection algorithm.
- Blockchain is a state-of-the-art technology that has great potential for wide deployment to ensure digital twin data security.
- As observed from Table IV-D, the existing research on integrating the blockchain and digital twin is trending. The main reason is that blockchain is a distributed database for digital twin and it can support other distributed AI algorithms, such as federated learning.

To study the AI-driven DT-IIoT practically, we look into the software tools for high-fidelity digital twin modeling and conduct a case study in Section V.

V. SOFTWARE TOOLS AND CASE STUDY OF DT-IIoT

A. Software Tools Development Roadmap

We use Fig. 14 to illustrate the development of software tools for IoT simulation which follows five representative periods from 2010 to 2025. Firstly, the simulation period starts before 2010 and ends after 2015, when many network simulators have been developed, such as NS-2/NS-3, OMNeT++, TOSSIM, Vienna Simulators, and others. Due to the lack of computing resources, those simulators usually mimic real networking scenarios with many assumptions. The simulation method has characteristics such as event-driven, simulated environment, non-interactive, and so on, which makes simulation results not accurate or applicable under many constraints

Secondly, the Hardware-in-the-loop (HIL) period starts after 2010 and ends before 2020. A few examples, such as the common open research emulator (CORE), extendable mobile ad hoc network emulator (EMANE), and others have been widely adopted to conduct modeling with real hardware and software. Compared to the simulation, HIL implements a real environment to conduct accurate modeling and conduct real-time task handling. For example, CORE has been adopted to test the performance of IoT protocols with both virtual

implementation and real devices, such as the constrained application protocol (CoAP) and others [194], [195].

Thirdly, the cross-platform co-simulation period starts after 2015 and ends before 2025. The cross-platform indicates the integration of different platforms, such as communication, computation, and control. The co-simulation is to conduct simulation jointly over multiple software with real-time synchronization. The co-simulation considers not only the impact of the communication, computation, and control components as a unified whole but also the complex interactions between them, which make the co-simulation realistic, compared to other simulation methods. The wireless cyber-physical simulator (WCPS), Simulink Real-Time™ and Simscape™, OPAL-RT simulator, and others are representative examples that have been vastly adopted for realistic modeling of the digital twin of IoT.

Fourthly, the digital twin period starts after 2015 and continues. The digital twin creates digital replicas of the IoT system, which can interact with the physical system continuously. The digital twin software tools integrate multiple software for data handling, analytics, management, visualization, and so on. The digital twin software is further developed based on cross-platform co-simulation tools. A few examples, such as the Bosch IoT suite, SAP Leonardo IoT Bridge, Azure Digital Twin, and others have been developed recently for digital twin modeling and testing.

Finally, the metaverse, as a universal and immersive virtual world, has been mentioned vastly and can be adapted for future software tools.

B. Open-source Software Tools for Digital Twin Modeling

Kamath *et al.* in [196] proposed open-source software tools for simulating the key functions of the integrated IIoT and digital twin, including data aggregation, data analytics, resource management, and visualization. Firstly, the Eclipse Hono is used to simulate the IoT network, which supports several interconnected IoT devices with common communication protocols, such as Message Queuing Telemetry Transport (MQTT), CoAP, and others. The Eclipse Hono implements common APIs to support uniform communication between devices using different protocols and applications. Secondly, Eclipse Ditto is adopted to implement the digital twin layer for IoT device management. Ditto provides mirroring between the physical world and the digital domain to enable off-the-shelf IoT functionalities. Ditto helps the digital twin to communicate efficiently and directly through APIs. Besides, Apache Kafka has integrated real-time data streaming processes. Apache Kafka provides scalable and distributed message queue service and ensures low latency, high concurrency, high reliability, and high throughput. Finally, Influx DB and Grafana are adopted for database data analytics and visualization services.

In their work, the industrial data, including the voltage, rotation, pressure, and vibration of industrial machines are adopted to validate the proposed framework. The experimental results demonstrated the effectiveness in the perspectives of robustness, dynamicity, and scalability.

Damjanovic-Behrendt *et al.* in [197] reviewed the open-source software for emulating the digital twin for smart

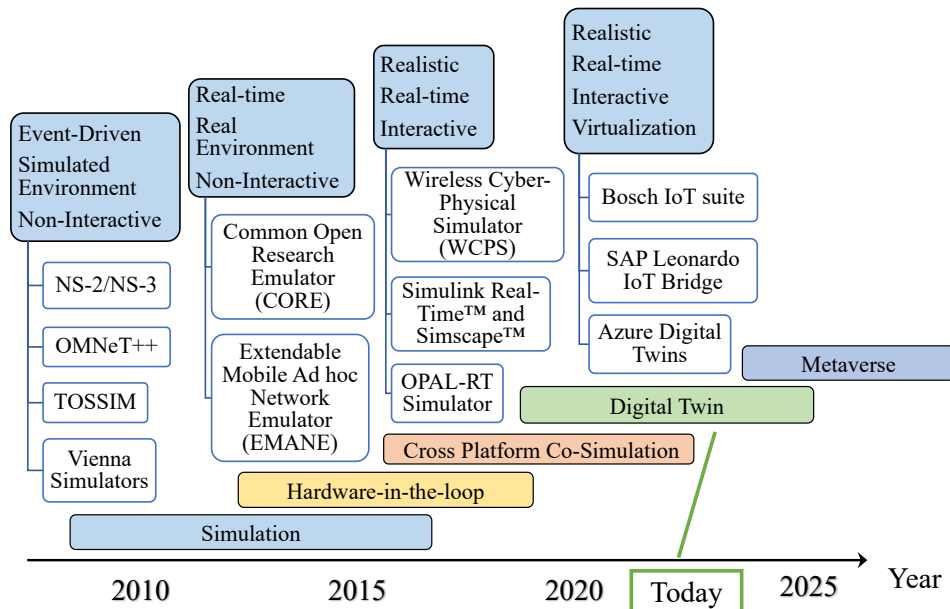


Fig. 14. The Development of Software Tools for IIoT

manufacturing, which can be categorized as data exchange software, middleware deployment tools, computational processing tools, machine learning software, and so on. The open-source software is developed by communities, such as Apache Software Foundation, the Linux Foundation, the Eclipse IoT, and so on, which vastly reduces the development cost.

To emulate the IoT data exchange, open-source data streaming tools, such as Apache Spark, Apache Storm, Apache Kafka, Apache Samza, Apache Flink, Apache Beam, and others. The data streaming tools are to handle data processing, such as data extraction, conversion, storage, and others that is one critical component in modeling digital twin systems.

The computational modeling tools are used for data aggregation, data analytics, and decision makings. The computational processing models can be classified as the batch data processing model and the time-series data processing tools, based on the data generation method. The batch data processing tools, such as Apache Hadoop, Apache HBase, and others are highly reliable, high-performance, scalable distributed storage systems. The time-series data processing tools are often used for industrial applications, such as InfluxDB, Elasticsearch, Logstash, Kibana (ELK) Stack, and others.

The middleware deployment tools include docker, Kubernetes, Eclipse Kapua, Eclipse Kura, and others that can be used to integrate and connect disparate software. Machine learning software is increasingly been adopted to drive the digital twin of intelligent decision-making, including TensorFlow, Keras, Caffe, Kaggle, PyTorch, and others. The seamless integration with data exchange software, middleware deployment tools, computational processing tools, and others is the key challenge of the digital twin software suite design.

To set an example, the authors proposed a demonstration framework, consisting of Apache Kafka as connecting APIs and data streaming pipeline, RabbitMQ for messaging services, Elasticsearch for data handling, and Apache Hadoop

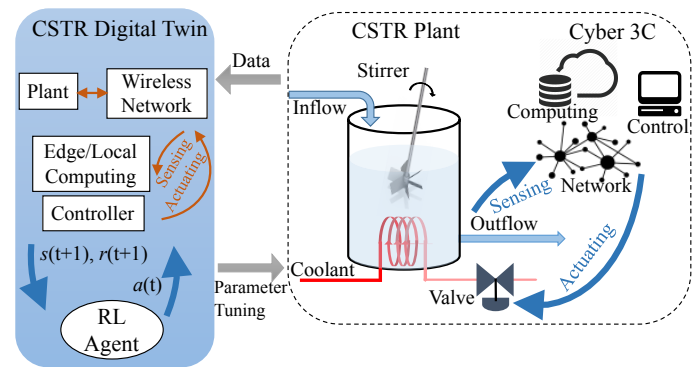


Fig. 15. The System Architecture of AI-driven DT-IIoT Case Study

and TensorFlow for data computation and analytics.

C. A Case Study of the Intelligent CSTR Digital Twin

Recall Section II-C1 that 3C is the critical virtual infrastructure of a fine-grained digital twin, which has not been integrated into existing digital twin models. In this section, we propose the first digital twin model with integrated 3C. We conduct a case study to illustrate the intelligent CSTR digital twin with integrated-3C, compared with the digital twin with sub-3C. Note that the integrated-3C considers all 3C subsystems and their interactions, whereas, the sub-3C only considers the 3C subsets.

D. Design Methodology

The CSTR digital twin is the digital counterpart of the CSTR that conducts two-way interactions, including receiving runtime system data and conducting parameter tunings with the CSTR. As observed from the system architecture in

Fig. 15, the CSTR consists of the interacting CSTR plant and cyber 3C. The CSTR plant is the reactor tank containing inflow for product input, outflow for effluent output, a stirrer to mix the product, and a coolant to maintain the tank temperature at a proper level to ensure the product quality and avoid over-heating. The CSTR plant interacts with the cyber 3C through sensing and actuating. The control component produces the control command that adjusts the coolant valve towards the setpoint temperature based on observations. The communication component transmits data between the temperature sensor, valve, and controller. The computing components provide the computing resource for calculating control commands.

The CSTR digital twin consists of the CSTR plant model and the cyber 3C, including the wireless network component, computing platform, and controller. The wireless network connects the plant and controller to enable sensing and actuating. Besides, towards the systematical and automated resource allocation, the reinforcement learning component in the CSTR digital twin interacts with CSTR to conduct parameter tunings based on the received runtime data.

The highlight of this case study is the consideration of the inherently intertwined cyber 3C. For example, the control component can increase the sampling rate to observe the plant frequently and obtain a better control performance. The high sampling rate also increases the data volume that needs to be delivered and poses a heavy traffic load to the wireless network. The noisy communication channel in factories and plants leads to severe packet drops, especially under network traffic. This degrades the control performance on the contrary. With the hybrid local/edge computing platform, the control tasks can be completed over the local controller or edge servers selectively. Thus, the hybrid local/edge computing platform can offload the control tasks to the local computers when the communication environment is severe. However, the edge server has unlimited computing resources compared to the local computer that can support computationally expensive controllers for better control performance.

Thus, it is necessary to carry out the integrated resource allocation and optimization approach with the consideration of all 3C components instead of their subset. To solve the integrated problem, the traditional mathematical model-based approach is impractical due to the complex interdependence between the 3C components. Thus, we leverage model-free reinforcement learning that can conduct parameter tunings on all 3C components through systematical trial and error.

E. CSTR Plant Modeling

The chemical process in the CSTR plant involves the irreversible exothermic reaction, which follows the material balance and energy balance. The CSTR material balance can be described as

$$\frac{dC_A}{dt} = \frac{q}{V}(C_{A_i} - C_A) - K_0 C_A \exp\left(-\frac{E}{RT}\right), \quad (1)$$

where C_A is product concentration in reactor mol/m^3 , the q is the inflow rate in m^3/s , V is the reactor volume in m^3 , C_{A_i} is the product concentration of inflow, K_0 , E , and R are constant values to calculate reaction rate, T is the reactor temperature.

The CSTR heat balance is described as

$$\frac{dT}{dt} = \frac{q}{V}(T_i - T) - \frac{\delta H K_0 C_A \exp\left(-\frac{E}{RT}\right)}{\rho C_p} + \frac{(T_{cs} - T)UA}{\rho C_p V}, \quad (2)$$

where UA is the heat transfer coefficient, δH is the reaction heat in J/mol , ρ is the product density in kg/m^3 , C_p is the heat capacity of product in J/kg-C , T_{cs} is the coolant temperature, T_i is the inflow temperature. We would like to tune the coolant temperature T_{cs} through the valve as shown in Fig. 15 to ensure the reactor temperature T remains a pre-defined value.

F. Q-learning Based 3C Integrated-Design in Digital Twin

We use the mean absolute error (MAE) to denote the control performance, which measures the distance between the current reactor temperature and setpoint value. The MAE is denoted as $\Delta T = \frac{1}{O} \sum_{i=1}^O (|T(i) - T_s(i)|)$, where O is the number of samples in control period, ΔT is the MAE, T_s is the setpoint temperature. Recall that 3C integrated design selects 3C parameters that are tightly related and carry out systematic trial and error in the digital twin until the optimal action sequence is found, i.e., optimal policy.

Thus, we formulate the 3C integrated-design problem as the Markov decision process (MDP), which has been adopted widely to model sequential decision-making problems. The MDP consists of four-tuple (X, A, P, R) . X is the state space. A is the action space. P is the transition probability. R is the reward function. The state is calculated by the MAE, such as $x(t) = \Delta T$, $x(t) \in X$. The action is defined as $a(t) = [a_{c1}(t), a_{c2}(t), a_{c3}(t)]$. $a_{c1}(t) \in [BPSK, OQPSK, ASK]$ is the communication parameter, such as the modulation types of IEEE 802.15.4-based industrial network. $a_{c2}(t) \in [localcomputer, edgeserver]$ denotes the selection of PID at local or MPC over the edge to generate control commands. $a_{c3}(t) \in [20Hz, 30Hz, 40Hz, \dots, 150Hz]$ is the sampling rate for control. The state transition probability $p(t)$ is unknown to the agent. Thus, we adopt model-free reinforcement learning, such as q-learning to carry out the trial and error in the digital twin. Finally, the reward function $r(t)$ is designed to assign a positive reward to the agent when the MAE is below a satisfactory threshold τ and vice versa.

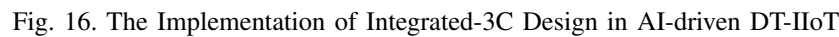
The agent is to find the optimal policy π^* that maximizes the expected cumulative reward:

$$\pi^* = \arg \max E\left(\sum_{j=0}^{\infty} \gamma^j r(t+j+1) | \pi\right), \quad (3)$$

where γ is the discount factor (e.g., 0.99), which discounts the future reward, and π is the policy.

The Q-learning approach creates a Q-table to store Q-values for each state-action pair, which denotes the estimated value of the state-action pair. At each time step t , the agent selects actions from the action set A based on the maximum Q-value. ϵ -greedy rule is applied to balance the exploration and exploitation. The Q-table updating process follows Equation 4.

$$Q(s(t), a(t)) \leftarrow Q(s(t), a(t)) + \alpha(r(t+1) + \gamma \max_a Q(s(t+1), a) - Q(s(t), a(t))), \quad (4)$$



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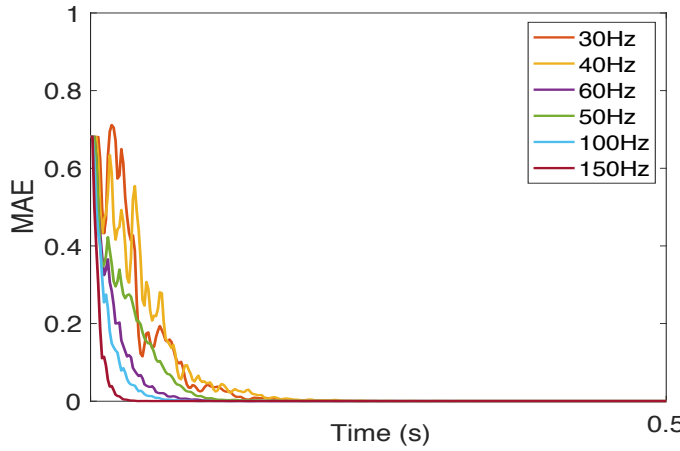


Fig. 18. Digital Twin with Sub-3C: MAE of MPC over Edge for the Integrated Control and Computing

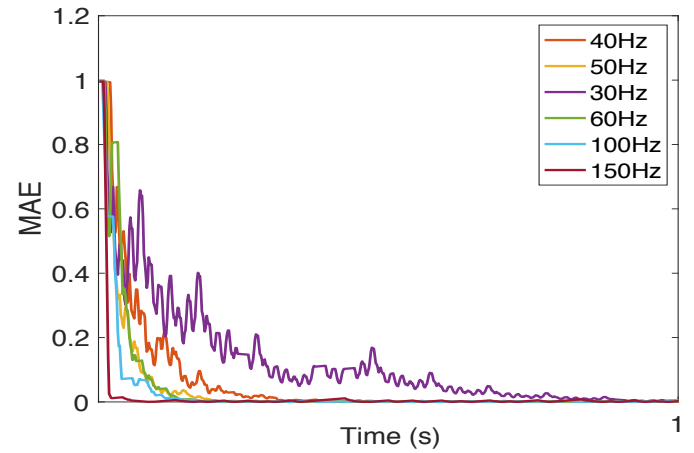
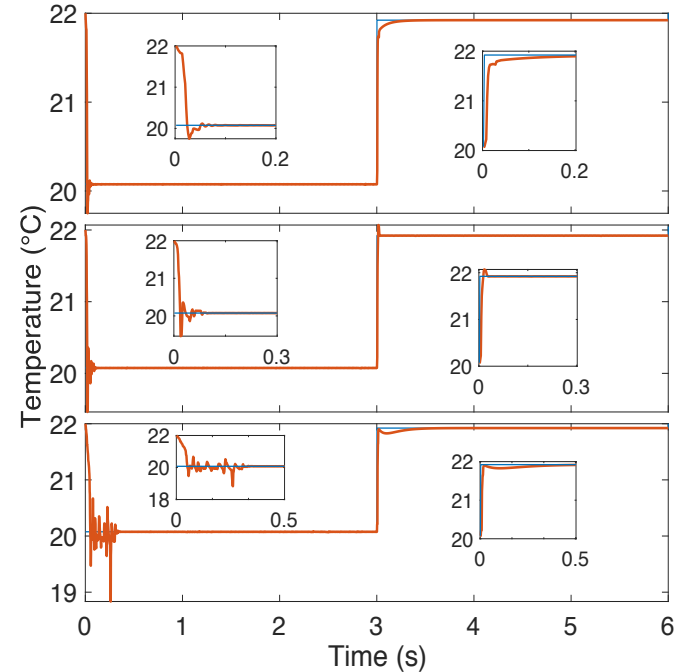


Fig. 19. Digital Twin with Sub-3C: MAE of MPC over Edge for the Integrated Control and Communication

and the selection on the computing platform as the key tuning parameters of the control and computing subsystems. The edge server that has sufficient computing resources can support powerful control algorithms, such as MPC. The increasing sampling rate, which conducts highly frequent sampling in response to physical disturbances is one control action to ensure the control performance. Without considering the impact of the communication subsystem, the MPC over the edge can conduct highly frequent sampling in response to physical disturbances quickly, where all the sampling data can be delivered successfully. As observed in Fig. 18, the control performance of MPC over edge increases with the increased sampling rate. The MAE of MPC over the edge decreases quickly from nearly 0.7 to 0 after around 0.2 seconds. For the higher sampling rate, such as 100Hz or 150Hz, the MAE of MPC over edge decreases faster than smaller sampling rates, such as 30Hz, or 40Hz, which indicates that the higher sampling rate brings better control performance assuming all packets can be delivered. The control performance of PID at local, remains the same under different sampling rates, as observed in Fig. 17. The main reason is that PID locally is a lightweight controller. The control performance of PID locally is limited and does not increase given a higher sampling rate.

3) *Scenario (iii)*: With the consideration of control and communication, the wireless network, which follows the IEEE 802.15.4 standard, connects the sensors, controllers, and actuators in the closed-loop [199]. The control system increases the sampling rate to sample the physical plant more frequently and improves the control performance. However, the high sampling rate results in a large number of data samples that are to be transmitted. The large data volume overloads the wireless network which has limited network resources and noisy industrial communication channels. This will result in severe packet drop especially when the industrial communication channels are noisy. Thus, the high sampling rate, on the contrary, could decrease control performance when considering network and control systems jointly.

As observed from Fig. 19, the increasing sampling rate increases the control performance of MPC over the edge. With



Mild, Medium and Heavy Noise Levels (Top to Bottom)

Fig. 20. Intelligent CSTR Digital Twin with Integrated-3C: Control, Communication, and Computing

the consideration of the communication subsystem, the packet drops due to the heavy traffic load of high sampling rates decrease the control performance, on the contrary, observed from the slow convergence at nearly 0.3 seconds. For the integrated control and communication case, the control performance of PID locally is the same as scenario (ii) in Fig. 17. This is because local computing is not affected by the packet drops.

4) *Scenario (iv)*: With the consideration of control, communication, and computing, Fig. 20 shows the Q-learning-based systematic parameter tuning for the integrated-3C. Q-learning conducts trial and error in the CSTR digital twin to decrease the error between the reactor temperature and setpoint. The errors at the first few iterations increase with

the noise levels go up, such as the mild, medium, and heavy noise levels. This is because the noise brings uncertainties to the system, which requires a longer time for the Q-learning algorithm to converge. As observed from the windows in Fig. 20, the control performance is converged in less than 0.2 seconds, 0.3 seconds, and 0.5 seconds under the mild, medium, and heavy noise level scenarios. In conclusion, Q-learning ensures the optimal control performance of the CSTR digital twin in the dynamic noisy industrial environment.

VI. FUTURE DIRECTIONS

In this section, we propose a few important future research directions, including the prospective applications, digital twin challenges, and integration with ABCDE.

A. Prospective Applications

Recall Table III that the digital twin has been an essential technology in many domains, including industry, transportation, energy, and so on. Some representative applications of the digital twin, such as smart manufacturing, product design, supply chain management, predictive maintenance, remote diagnosis, and many others have been proposed and well-studied. In this section, we discuss two prospective applications based on our in-depth investigation.

1) *“virtual-cyber-physical” Integrated Resource Management*: Resource allocation for communication networks, which includes spectrum management, link scheduling, routing, and so on greatly impacts packet loss, delay, and throughput [200]. Computing resources, such as CPU, storage, etc, shall be managed effectively to improve the completion of computing tasks [199]. Design on the sampling, estimation, and control strategies, such as event-triggered schemes, and self-triggered schemes can maximize the quality of control [201]. The integrated design synthesizes communication, computing, and control that are interrelated inherently and is foreseen to be a valuable but challenging research problem.

To solve the integrated design problem, the digital twin, which interconnects the virtual, cyber, and physical domains is designed to be the systematical and practical experiment platform that can carry out integrated resource management. The integrated resource management through the cross “virtual-cyber-physical” digital twin is an ideal application scenario. However, the research efforts in this direction are inadequate and need further studies.

2) *Monitoring, Analysis, Modeling, Prediction, and Control for the Integrated System Security*: Cyber attacks against smart grids, power plants, oil and gas pipelines, and other smart IIoT systems are diversified, stealthy, and destructive [202], [203]. Due to the massive interactions between the cyber domain and critical infrastructures, smart IIoT systems suffer from extendable vulnerabilities that are different from traditional networking systems. For example, the smart IIoT systems have a broader attack surface, including cyber domains, such as computing, communication, and control, and physical domains, such as controller, sensor, and actuators, and cross-domain destructiveness, such as the disruption of the physical

processes due to the malicious cyber operations, and significant economic losses due to the adversarial actions in real-world [204], [205]. Thus, the system security desires adequate monitoring, analysis, modeling, prediction, and control mechanisms that enable continuous monitoring and control, cross-domain and holistic analysis and modeling, and proactive and predictive risk assessment and mitigation.

To establish the above security mechanisms, the digital twin technology, which establishes the virtual representations and conducts simultaneous synchronization with smart IIoT systems can be of great use. In specific, the digital twin interacts with the smart IIoT system by collecting data and sending command signals. The data reflect the system operating conditions and enable situational awareness of insecure events. The command signals are the operational actions to ensure secure operations. The digital twin can carry out cross-domain and holistic analysis based on the collected data and high-precision smart IIoT system models. Empowered by AI algorithms, the digital twin can conduct a timely assessment of the security risks based on historical data and runtime data, and carry out risk mitigation strategies. Yet, it is still a trending research topic that leverages the digital twin for integrated system security.

B. Digital Twin Challenges

Based on our survey, we outline the challenges of digital twins, such as real-time and high-precision digital twins, “general” digital twins, and self-evolving and predictive digital twins.

1) *Real-time and High-precision Digital Twin*: From a digital twin perspective, real-time data exchange with real-world smart IIoT systems is an essential attribute to achieve timely situational awareness. However, existing data communication infrastructures cannot support large amounts of data exchange with guaranteed latency. Therefore, real-time data exchange seeks advanced communication networks, such as 6G [206]. Recall that in Section II-B1, the digital twin must be fine-grained, which collects not only real data but also interrelated 3C processes for the correctness and accuracy of the command signals generated in the digital twin. Therefore, based on our survey, we expect that future digital twins must hold real-time and high-precision characteristics.

2) *“General” Digital Twin*: Smart IIoT systems, such as smart factories, are characterized by large scale, wide geographic distribution, and heavy assets [90]. Digital twins are also considered to be the most valuable virtual assets of an organization. However, for small-scale organizations, it is costly and inefficient to build high-quality digital twins. For a single organization in an application domain, it is costless to build similar digital twins repeatedly. Therefore, it is necessary to develop a “General” digital twin that contains only the basic infrastructure of an intelligent IIoT system, such as control, communication, and computation. Other organizations can reuse the “General” digital twin by simply fine-tuning the parameters of the basic infrastructure, such as network settings, control algorithms, and computation methods, and injecting its data. In this way, the “General” digital twin

avoids duplicate digital twin development and improves overall efficiency.

3) *Self-evolving and Predictive Digital Twin*: Digital twins, containing large amounts of data, highly accurate system models, and customized virtual infrastructure, are the most valuable virtual assets. In the context of constant change, manually reconfiguring and updating a large-scale digital twin is expensive and inappropriate. The self-evolving and predictive nature of digital twins suggests that digital twins can reconfigure and update themselves predictively in response to changing system settings without manual intervention [35]. Recall the review in Section III, AI algorithms address the problem of prediction, classification, and automation in digital twins. Based on the massive amount of data, high-precision system models, and customized virtual infrastructure in digital twins, AI algorithms such as reinforcement learning can be used to achieve self-evolving and predictive digital twins.

C. Integrating with ABCDE

Technology paradigms such as artificial intelligence, blockchain, cloud computing, big data, and edge computing have significantly improved the efficiency, capabilities, and security of computer-based systems over the past few years [207]–[210]. The digital twin is considered to be the emerging paradigm that creates virtual domains for computer-based systems and further improves intelligence, efficiency, and security. Based on our survey, we outline possible future research directions for the integration of digital twins with ABCDE.

1) *Integration of AI and Digital Twin*: AI paradigms, such as neural networks, deep learning, and reinforcement learning, have been valuable research topics in the past decades. Dozens of AI algorithms, such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Deep Q Networks (DQN), have been invented to solve many real-world critical problems, such as natural language processing, autonomous driving, etc. Digital twins play an important role in connecting human intelligence and artificially intelligent machine intelligence to enable intelligent decision making in the real world. Several research efforts have investigated the promising integration of AI and digital twins, such as [109], [211]. However, it is still an emerging research topic that deserves input research efforts.

2) *Integration of Blockchain and Digital Twin*: Blockchain is an innovative paradigm for distributed data storage where blocks of data are linked to ensure data security. Blockchain has been adopted to provide an environment of trust and security for collaboration between multiple participants. Digital twins are the most valuable virtual assets, exchanging data with real-world entities at scale throughout the application lifecycle. Therefore, a secure data management model for complex digital twins of interactive digital twins and real-world participants of intelligent IIoT systems is necessary. There are research papers using blockchain to make digital twins [166], [189], [212]. However, for many smart IIoT systems, the need to study the integration of blockchain and digital twins is significant.

3) *Integration of Cloud and Digital Twin*: Cloud computing is a proven computing technology that provides nearly unlimited, on-demand computing resources. Cloud-based digital twins utilize cloud servers for large-scale simultaneous processing of digital twins, such as monitoring, evaluation, and control. The integration of cloud and digital twin has been adopted in many application areas such as industrial [213], [214], medical [111], and transportation [100]. The resource-consuming tasks of the digital twin can be assigned to the cloud for faster computational services, while the communication distance and the latency caused by large amounts of data may degrade the quality of service. The tradeoff between computational resources and communication latency is a key research question for the integration of cloud computing and digital twins and needs further study.

4) *Integration of big Data and Digital Twin*: Big data denotes a large amount of data that is collected from the production lifecycle, including structured, semi-structured or unstructured data sets. Data mining techniques, such as statistical modeling and machine learning, analyze and extract useful information from the massive amounts of data. The integration of big data and digital twins enables big data-driven management, diagnostic and maintenance solutions. Existing research work provides feasibility analysis and architecture design for manufacturing [12], [215]. The demand for further investigation into the integration of big data and digital twin is significant.

5) *Integration of Edge computing and Digital Twin*: Edge computing leverages the computing resources of local edge devices, such as routers and switches, to avoid the communication latency of sending large amounts of data to cloud servers. Pushing computational tasks to the edge can significantly reduce communication latency, which is beneficial for time-sensitive applications such as Cardio Twin, a digital twin of the human heart [49]. For real-time applications, such as manufacturing, healthcare, etc., the integration of edge computing and digital twins has great potential for widespread adoption, which has yet to be extensively studied.

6) *The Cross-Integration of ABCDE and Digital Twin*: The cross-integration of ABCDE and digital twin includes possible mix-match combinations between ABCDE, such as AB-DT, ABC-DT, ABCD-DT, ABCDE-DT, BC-DT, BCD-DT, and so on. For example, CE-DT is the integration of cloud computing, edge computing, and the digital twin, such as Qi *et al.* in [214] discussed the smart IIoT systems and DT supported by cloud computing and edge computing. The ABE-DT indicates the cross-integration of AI, blockchain, edge computing, and digital twin, such as Lu *et al.* in [32] studied the federated learning and blockchain for the digital twin-based edge networks. In addition, Nguyen *et al.* in [209] investigated federated learning and blockchain in mobile edge computing (MEC). The cross-integration of ABCDE and the digital twin remains an open research direction, which demands greater research efforts.

VII. CONCLUSION

The DT-IIoT leverages the digital twin, as key strategic information technology, to empower the intelligence and security of IIoT systems. To systematically survey the existing

research on the digital twin, we reviewed preliminaries, real-world applications, and digital twin architecture and models. We investigated not only the classic machine learning solutions for prediction, classification, and automation problems of IIoT but also the state-of-the-art AI Solutions, such as transfer learning and federated learning. Besides, we investigated data security and control security issues, as well as state-of-the-art blockchain-based security solutions. Moreover, software tools for high-fidelity digital twin modeling are discussed. We conducted a case study on AI-driven DT-IIoT, which consists of the CSTR process control system for validation and reinforcement learning-based integrated control and computing resource allocation. Finally, digital twin prospective applications, challenges, and integration with ABCDE are outlined.

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